Electromagnetic Field Theory

Classroom Workbook

ELECTRONICS AND COMM. ENGINEERING

ELECTRICAL ENGINEERING

INSTRUMENTATION ENGINEERING

GATE/ESE

unacademy

Electromagnetic Field Theory

Classroom Workbook

For ECE, EE, EI

Copyrights © All Rights Reserved

Sorting HAT Technologies Pvt. Ltd.

No part of this publication may be reproduced or distributed in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise or stored in a database or retrieval system without the prior written permission of the publishers. The program listings (if any) may be entered, stored and executed in a computer system, but they may not be reproduced for publication.

Printing of books passes through many stages - writing, composing, proof reading, printing etc. We try our level best to make the book error- free. If any mistake has inadvertently crept in, we regret it and would be deeply indebted to those who point it out. We do not take any legal responsibility.

Edition : EMFT-2024/2025

Sorting HAT Technologies Pvt. Ltd.

353, 12th 'A'cross, off double road, Opp. City Centre

Residency, Indira nagar, 1st stage Bangalore

Bangalore KA560038IN.

GATE Syllabus

ECE

Maxwell's equations: differential and integral forms and their interpretation, boundary conditions, wave equation, Poynting vector. Plane waves and properties: reflection and refraction, polarization, phase and group velocity, propagation through various media, skin depth. Transmission lines: equations, characteristic impedance, impedance matching, impedance transformation, S-parameters, Smith chart. Rectangular and circular waveguides, light propagation in optical fibers, dipole and monopole antennas, linear antenna arrays.

EE

Coulomb's Law, Electric Field Intensity, Electric Flux Density, Gauss's Law, Divergence, Electric field and potential due to point, line, plane and spherical charge distributions, Effect of dielectric medium, Capacitance of simple configurations, Biot-Savart's law, Ampere's law, Curl, Faraday's law, Lorentz force, Inductance, Magnetomotive force, Reluctance, Magnetic circuits, Self and Mutual inductance of simple configurations.

IN

Coulomb's Law, Electric Field Intensity, Electric Flux Density, Gauss's Law, Divergence, Electric field and potential due to point, line, plane and spherical charge distributions, Effect of dielectric medium, Capacitance of simple configurations, Biot-Savart's law, Ampere's law, Curl, Faraday's law, Lorentz force, Inductance, Magnetomotive force, Reluctance, Magnetic circuits, Self and Mutual inductance of simple configurations.

ESE Syllabus

E & T

Elements of vector calculus, Maxwell's equations—basic concepts; Gauss' and Stokes' theorems; Wave propagation through different media; Transmission lines—different types, basics, Smith's chart, impedance matching and transformation, S-parameters, pulse excitation, uses; Waveguides—basics, rectangular types, modes, cut-off frequency, dispersion, dielectric types; Antennas—radiation pattern, monopoles/dipoles, gain, arrays—active/passive, theory, uses.

EE

Line Plane and Spherical Charging Distribution, Ampere's and Bio-Sava RT's Law. Maxwell Equation

Contents

Sr. Chapters

1. Coordinate System and Electrostatics

2. Magnetostatics

3. Plane Wave Propagation

4. Transmission Lines

5. Waveguides

6. Antennas

7. Optical Fibres

■■■■■

1 Coordinate System and Electrostatics

MCQ, MSQ and NAT

Q.1 The direction of vector **A** is radially outward from the origin, with $|A| = kr^n$ where $r^2 = r^2 + y^2 + z^2$ and k is a constant. The value of n for which $\nabla \cdot A = 0$ is

(A) –2 (B) 2

(C) 1 (D) 0

Q.2 Consider a vector field $\vec{A}(\vec{r})$. The closed loop line integral $\oint \vec{A} \cdot d\vec{l}$ can be expressed as

(A) $\iint (\nabla \times \vec{A}) \cdot d\vec{s}$ over the closed surface bounded by the loop

(B) $\iiint (\nabla \cdot \vec{A}) dv$ over the closed volume bounded by the loop

(C) $\iiint (\nabla \cdot \vec{A}) dv$ over the open volume bounded by the loop

(D) $\iint (\nabla \cdot \vec{A}) d\vec{s}$ over the open surface bounded by the loop

Q.3 The divergence of vector field $\vec{A} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$ is

(A) 0 (B) 1/3

(C) 1 (D) 3

Q.4 The force on a point charge + q kept at a distance d from the surface of an infinite grounded metal plate in a medium of permittivity $\in$ is

(A) 0

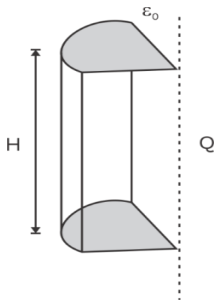
(B) $\frac{q^2}{16\pi \in d^2}$ away from the plate

(C) $\frac{q^2}{16\pi \in d^2}$ towards the plate

(D) $\frac{q^2}{4\pi \in d^2}$ towards the plate

Q.5 Given the vector $A=(\cos x)(\sin y)\hat{a}_x+(\sin x)(\cos y)\hat{a}_y$, where $\hat{a}_x, \hat{a}_y$ denote unit vector along x,y directions, respectively. The magnitude of curl of A is _____.

Q.6 What is the electric flux $\left(\int \vec{E}.d\hat{a}\right)$ through a quarter-cylinder of height H (as shown in the figure) due to an infinitely long line charge along the axis of the cylinder with a charge density of Q?



(A) $\frac{4H}{Q\varepsilon_0}$ (B) $\frac{HQ}{4\varepsilon_0}$

(C) $\frac{HQ}{\varepsilon_0}$ (D) $\frac{H\varepsilon_0}{4Q}$

Q.7 For a vector field $\vec{A}$, which one of the following is FALSE?

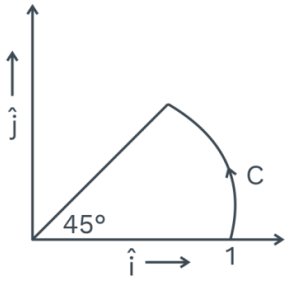
(A) $\vec{A}$ is solenoidal if $\nabla . \vec{A} = 0$.

(B) $\nabla \times \vec{A}$ is another vector field.

(C) $\vec{A}$ is irrotational if $\nabla^2 \vec{A} = 0$.

(D) $\nabla \times \left(\nabla \times \vec{A} \right) = \nabla \left(\nabla . \vec{A} \right) - \nabla^2 \vec{A}$

Q.8 The vector function $\vec{F}(r) = -x\hat{i} + y\hat{j}$ is defined over a circular arc C shown in the figure.



The line integral of $\int_C F(r) \cdot dr$ is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{1}{6}$ (D) $\frac{1}{3}$

Q.9 If the vectors (1.0, –1.0, 2.0), (7.0, 3.0, x) and (2.0, 3.0, 1.0) in R^3 are linearly dependent, the value of x is _____.

Q.10 Consider the vector field $\vec{F} = \hat{a}_x (4y - c_1 z) + \hat{a}_y (4x + 2z) + \hat{a}_z (2y + z)$ in a rectangular coordinate system (x, y, z) with unit vectors $\hat{a}_x$, $\hat{a}_y$, and $\hat{a}_z$. If the field $\vec{F}$ is irrotational (conservative), then the constant c_1 (in integer) is _____

Q.11 For a vector field $D = \rho \cos^2\varphi \, a_\rho + z^2 \sin^2\varphi \, a_\varphi$ in a cylindrical coordinate system (ρ, φ, z) with unit vectors a_ρ, a_φ and a_z, the net flux of D leaving the closed surface of the cylinder ($\rho = 3$, $0 \le z \le 2$) (rounded off to two decimal places) is _____

Q.12 Consider two position vectors $\vec{r}_1 = 5\hat{a}_x + 3\hat{a}_y + 2\hat{a}_z$ and $\vec{r}_2 = 3\hat{a}_x + 2\hat{a}_y + 3\hat{a}_z$, both expressed in centimeter units. Determine the unit vector pointing from $\vec{r}_1$ *to* $\vec{r}_2$ and the distance between these two given position vectors.

(A) $4.9\hat{a}_x - 2.45\hat{a}_y + 2.45\hat{a}_z; \sqrt{6}$ cm

(B) $2.45\hat{a}_x + 2.45\hat{a}_y - 4.9\hat{a}_z; 6$ cm

(C) $-0.8\hat{a}_x - 0.4\hat{a}_y + 0.4\hat{a}_z; \sqrt{6} cm$

(D) $-2\hat{a}_x - \hat{a}_y - \hat{a}_z; \sqrt{3} cm$

Q.13 A charge of magnitude 4×10^{-4} C is located at M(2, 2, 2), and another charge of magnitude -2×10^{-4} C is located at N(2, 0, 5) in a

vacuum. Determine the force exerted on Q_2 by Q_1 if both the charges are placed in a vacuum.

[Given, $\epsilon_0 = 8.854 \times 10^{-12} \simeq \frac{10^{-9}}{36\pi}$ F/m]

(A)$(20.72\hat{a}_x - 46.05\hat{a}_z)\ N$

(B)$(30.72\hat{a}_y - 46.08\hat{a}_z)\ N$

(C)$(25\hat{a}_x - 30.72\hat{a}_y - 46.08\hat{a}_z)\ N$

(D)$(4\hat{a}_x + 30.72\hat{a}_y + 2\hat{a}_z)\ N$

Q.14 A region (z<0) is characterized by $\epsilon_{r2} = 2$ and the other region (z > 0) is characterized by $\epsilon_{r1} = 5$. If $\vec{D}_1 = 5\hat{a}_x + 2\hat{a}_y - 3\hat{a}_z nC/m^2$ and no charge exists at the interface, then $\overrightarrow{D}_2$ will be-

(A)$2\hat{a}_x + 0.8\hat{a}_y\ nC/m^2$

(B)$-2\hat{a}_x - 0.8\hat{a}_y + 3\hat{a}_z\ nC/m^2$

(C)$0.8\hat{a}_y - 3\hat{a}_z\ nC/m^2$

(D)$2\hat{a}_x + 0.8\hat{a}_y - 3\hat{a}_z\ nC/m^2$

Q.15 Consider a parallel plate capacitor that is maintained at 300 Volts. If the separation between the plates of the capacitor and the area of the plates is 5 mm and 20 cm^2 respectively, then the displacement current is _____(in mA). (Up to three decimal places)

[Consider t = $1\ \mu s$ and $\epsilon_0 = 8.85 \times 10^{-12}$ F/m]

Q.16 The angle between two position vectors $(30\hat{a}_x + 20\hat{a}_y - 10\hat{a}_z)\ m$ and $(30\hat{a}_x - 20\hat{a}_y + 20\hat{a}_z)\ m$ is_____ _____(in degree). (Up to two decimal places)

Q.17 Three points are given in the cartesian coordinate system as P (4, –3, 2), Q(–3, –2, 6), and R(3, 6, –4). Determine the vector from point P to the mid-point of the straight-line joining Q to R.

(A)$4\hat{a}_x - 5\hat{a}_y + \hat{a}_z$

(B)$-4\hat{a}_x + 5\hat{a}_y - \hat{a}_z$

(C)$4\hat{a}_x - 3\hat{a}_y + 2\hat{a}_z$

(D)$-3\hat{a}_x - 2\hat{a}_y + 6\hat{a}_z$

Q.18 A charge is distributed uniformly over the plane z = 10 cm with a density of $\left(\frac{1}{3\pi}\right)\ \mathrm{nC/m^2}$. The magnitude of $\vec{E}$ (in V/m) for z > 10 cm is_____ (Rounded off to the nearest integer)

$\left[Consider\ \epsilon_0 = \left(\frac{10^{-9}}{36\pi}\right)\ \mathrm{F/m}\right]$

Q.19 A scalar field is given as-

T = 25rsinϕ (In cylindrical coordinate system)

Determine the gradient at $U\left(\sqrt{2}, \frac{\pi}{2}, 5\right)$.

(A)$25\hat{a}_r$ (B)$25\hat{a}_\phi$

(C)$5\hat{a}_r$ (D)$25\hat{a}_z$

Q.20 Consider that a cube has sides of 2 m and its center is placed at (0,0,0). The sides of the cube are parallel to the coordinate axes. If the electric flux

density is $x^3\hat{a}_x + x^2y^2\hat{a}_z$ C/m^2, then the charge inside the cube is-

(A)4 C (B)8 C

(C)16 C (D)32 C

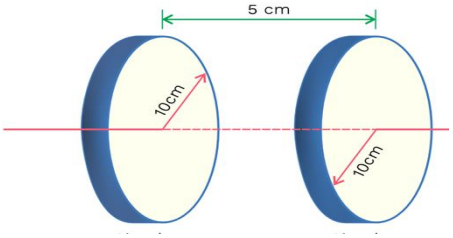
Q.21 The electric field intensity at point x = 0 due to a set of infinite number of point charges each equal to 'q' placed in a medium with permittivity ϵ and along x-axis at x = 1, x = 2, x = 4, x = 8 and so on is-

(A)$\frac{q}{34\pi\epsilon}$ (B)$\frac{q}{3\pi\epsilon}$

(C)$\frac{q}{\pi\epsilon}$ (D)$\frac{q}{2\pi\epsilon}$

Q.22 A capacitor is made up of two circular plates, each of radius 10 cm and separated by a distance of 5 cm as shown in the figure below. Assume that the capacitor is being charged by an external source. The charging current is constant and equal to 0.20 Amperes. Which of the given options is/are correct?

[Assume $\in_0$ = 8.854 × 10^{-12} F/m]



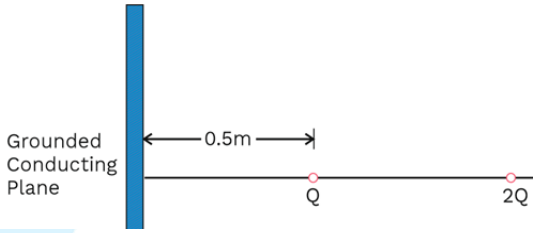
(A)The change in the potential difference between the plates with respect to time is 8.87 × 10^9 V/s

(B)The capacitance between the plates is 80.321 pF.

(C)The change in the potential difference between the plates with respect to time is 35.95 × 10^9 V/s.

(D)The capacitance between the plates is 5.563 pF.

Q.23 Consider that the two point charges Q and 2Q are placed along the x-axis in front of a grounded, infinite conducting plane as illustrated in the figure. The force acting on the charge Q is-



(A)$\frac{2Q^2}{9\pi\epsilon_0}$ N (B)$\frac{2Q^2}{4\pi\epsilon_0}$ N

(C)$\frac{7Q^2}{8\pi\epsilon_0}$ N (D)$\frac{5Q^2}{4\pi\epsilon_0}N$

Q.24 Consider 4-point charges placed in free space at points in Cartesian coordinates (-5, 0, 0), (5, 0, 0), (0, -5, 0) and (0, 5, 0). The magnitude of four equal charges is 10 µC. The force on another charge of magnitude 25 µC placed at (0, 0, 10) will be _____ (N). [Up to 2 decimal places]

$[Consider\ \varepsilon_0 = 8.85 \times 10^{-12}\ \mathrm{F/m}]$

[Assume all given distances are in centimeters]

Q.25 Assume two infinite lines of charge each carrying a charge density (ρ_L) are parallel to the z-axis and are located at x =1 and x =-1.

The $\overrightarrow{E}$ at an arbitrary point in free space along the y-axis will be –

(A) $\frac{y \cdot \rho_L}{2\pi\epsilon_0 (1+y^2)}\ \hat{a}_y$

(B) $\frac{\rho_L}{\pi\epsilon_0 (1+y^2)}\ \hat{a}_y$

(C) $\frac{2y\rho_L}{\pi\epsilon_0\sqrt{1+y^2}}\hat{a}_y$

(D) $\frac{y\rho_L}{\pi\epsilon_0 (1+y^2)}\hat{a}_y$

Q.26 A uniform line charge, infinite in extent, with $\rho_L = 20\text{nC/m}$ lies along the -axis. $\vec{E}$ at $(5, 12, 3)\text{m}$ is-

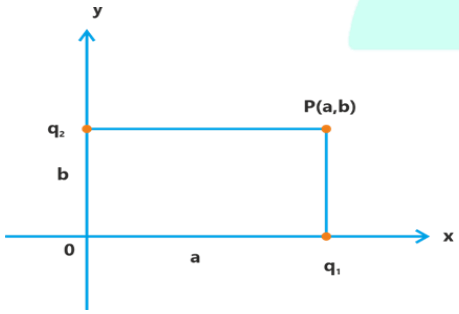
(A) $138.45\hat{a}_x + 332.28\hat{a}_y\ \text{V/m}$

(B) $332.281\hat{a}_x + 138.45\hat{a}_y\ \text{V/m}$

(C) $138.45\hat{a}_x - 332.28\hat{a}_y\ \text{V/m}$

(D) $332.281\hat{a}_x + 138.45\hat{a}_y\ \text{V/m}$

Q.27 Two point charges q_1 and q_2 are placed at a distance a = 3 cm and b = 2 cm from the origin as shown in the figure below.



If q_1 = 4 μC and q_2 = 1 μC, then the net electric field vector at point P is-

[Assume Coulomb's constant, $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \text{Nm}^2/\text{C}^2$]

(A) $4 \times 10^7\hat{a}_y + 2.25 \times 10^7\hat{a}_x$

(B) $4 \times 10^7\hat{a}_x + 2.25 \times 10^7\hat{a}_y$

(C) $4 \times 10^7\hat{a}_x - 2.25 \times 10^7\hat{a}_y$

(D) $4 \times 10^7\hat{a}_y - 2.25 \times 10^7\hat{a}_x$

Q.28 Three vectors are given as-

$\overrightarrow{A} = 2\hat{a}_x + \hat{a}_y$

$\vec{B} = 3\hat{a}_x + 2\hat{a}_y$

$\vec{C} = 5\hat{a}_y + \hat{a}_z$

Find the cross product $(\vec{A} \times \vec{B}) \times \vec{C}$

(A) $11\hat{a}_x + 2\hat{a}_y + 10\hat{a}_z$

(B) $11\hat{a}_x - 2\hat{a}_y + 10\hat{a}_z$

(C) $19\hat{a}_x + 2\hat{a}_y + 10\hat{a}_z$

(D) $19\hat{a}_x - 2\hat{a}_y + 10\hat{a}_z$

Q.29 An electric potential is given as V = $5x-7x^2y + 8y^2 + 16yz – 5z$. The electric field at (2, 4, -3) is-

(A) $107\hat{a}_x + 12\hat{a}_y - 59\hat{a}_z$

(B) $107\hat{a}_x - 12\hat{a}_y - 59\hat{a}_z$

(C) $107\hat{a}_x + 12\hat{a}_y + 59\hat{a}_z$

(D) $107\hat{a}_x - 12\hat{a}_y - 59\hat{a}_z$

Q.30 Find the electric field at point P(6, -2, 4) due to two charges, q_1 = 3µC at point (1, 1, 1) and q_2 = $2\mu\text{C}$ at point (-2, 4, 6).

[Assume Coulomb's constant, $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \text{Nm}^2/\text{C}^2$.]

(A $614.51\hat{a}_x - 389.07\hat{a}_y + 253.31\hat{a}_z$

(B) $614.51\hat{a}_x + 389.07\hat{a}_y + 253.31\hat{a}_z$

(C) $342.99\hat{a}_x - 185.43\hat{a}_y - 321.19\hat{a}_z$

(D $342.99\hat{a}_x + 185.43\hat{a}_y + 321.19\hat{a}_z$

Q.31 A parallel plate capacitor is having an area of 25 cm^2 & distance between plates d = 10 mm with $\varepsilon = 2\varepsilon_0$. If the voltage applied across the plates is 50

sin100t, then the displacement current is-

(A)22.125 sin100t nA

(B)22.125 sin100t A

(C)22.125 cos100t A

(D)22.125 cos 100t nA

Q.32 If $\vec{A} = 3\hat{a}_y + 4\hat{a}_z$ and $\vec{B} = 4\hat{a}_y - 10\hat{a}_z + 5\hat{a}_z$, then the unit vector perpendicular to both $\vec{A}$ and $\vec{B}$

(A)$\frac{1}{\sqrt{3425}}(-55\hat{a}_x + 16\hat{a}_y - 12\hat{a}_z)$

(B)$\frac{1}{58.52}(55\hat{a}_x - 16\hat{a}_y - 12\hat{a}_z)$

(C)$\frac{1}{58.52}(55\hat{a}_x + 16\hat{a}_y - 12\hat{a}_z)$

(D)$\frac{1}{\sqrt{3425}}(55\hat{a}_x + 16\hat{a}_y + 12\hat{a}_z)$

Q.33 The magnitude of the force on a charge 30 μC placed at (0, 0, 2) m due to another charge -200 μC located at (2, 1, 0) m is-

[Consider $\epsilon_0 = \left(\frac{10^{-9}}{36\pi}\right)$ F/m].

(A)6 N (B)4 N

(C)2 N (D)9 N

Q.34 Two-point charges 8 nC and -8nC are separated by a distance of 0.4 m. The energy in the system of these two charges is _____nJ. (Rounded off to the nearest integer).

[Consider $\epsilon_0 = 8.85 \times 10^{-12} F/m$].

Q.35 A parallel-plate capacitor contains two dielectrics, $\varepsilon_{r1} = 2$ and $\varepsilon_{r2} = 3.3$. If the area of the plates is 2 m^2 and the distance between the plates is 10^{-3} m, then the capacitance of the parallel-plate capacitor is _____ nF. (Up to 2 decimal places).

[Consider that both the dielectrics cover half of the volume each and $\epsilon_0 = 8.85 \times 10^{-12}$ F/m]

Q.36 If the electric flux density is given as $\overrightarrow{D} = 2x^3\hat{a}_x + x^2y\hat{a}_y + x^3y^3\hat{a}_z$, then the charge enclosed inside the region -1 < x < 1, -1 < y < 1 and -2 < z <2 is _____ C. (Up to 2 decimal places).

[Consider the dimensions in meters].

Q.37 If the electric field vector is $\overrightarrow{E} = 3\hat{a}_x - 5\hat{a}_y + 2\hat{a}_z$ and the magnetic field vector is $\overrightarrow{H} = 2\hat{a}_x + 4\hat{a}_y - 5\hat{a}_z$, then, $\overrightarrow{E} \times \overrightarrow{H}$

(A)$2\hat{a}_x - 19\hat{a}_y + 22\hat{a}_z$

(B)$17\hat{a}_x + 19\hat{a}_y + 22\hat{a}_z$

(C)$2\hat{a}_x + 19\hat{a}_y + 2\hat{a}_z$

(D)$17\hat{a}_x - 19\hat{a}_y + 22\hat{a}_z$

Q.38 Match List-I answer using with List-1l and select the correct | the codes given below the lists:

List-I		List-II	
A.	The divergence of a vector field is zero	1.	Conservative field
B.	The curl of a vector field is zero	2.	Solenoidal
C.	The dot product of two vector fields is zero	3.	Irrotational
D.	The line integral of static electric field around closed path is zero	4.	Perpendicular

(A)A-1 B-2 C-3 D-4

(B) A-3 B-4 C-1 D-2

(C)A-4 B-1 C-2 D-3

(D)A-2 B-3 C-4 D-1

Q.39 The angle between two vector, $\vec{A} = 3a_x + 6a_y \text{ and } \vec{B} = 5a_y - 3a_z \; is$

(A)44°

(B)53.7°

(C)37.3°

(D)39.92°

Q.40 The electric field for a given charge enclosed in a spherical gaussian surface is E. What is the electric field if charge enclosed and radius of gaussian surface is doubled?

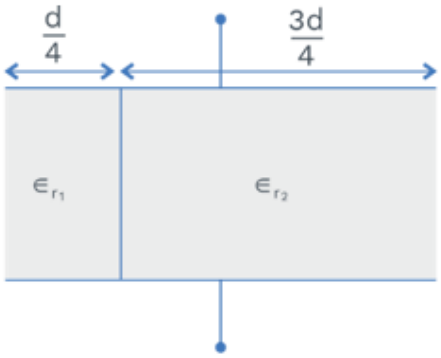
(A)Electric field is 2 times

(B)Electric field is same

(C)Electric field is half

(D)Electric field is 4 times

Q.41 An air filed parallel-plate capacitor with capacitance C. If it is filled with dielectric material as shown in figure.



The new capacitance is _____.

(A) $C\left(4\epsilon_{r_1} + \frac{4}{3}\epsilon_{r_2}\right)$

(B) $C\left[\frac{4}{3}\epsilon_{r_1} + 4\epsilon_{r_2}\right]$

(C) $\frac{C}{3}\left[4\epsilon_{r_1} + \epsilon_{r_2}\right]$

(D) $\frac{C}{3}\left[\epsilon_{r_1} + 12\epsilon_{r_2}\right]$

Q.42 Which of the following statement(s) is/are correct:

1. Gradient is an operation performed on a scalar function which results in a vector function

2. The Laplacian of a scalar field **V** is the divergence of the gradient of **V** and for a scalar quantity, it results in a vector.

3. The Laplacian of vector '**A**' is the gradient of divergence of **A** minus the curl of **A**.

(A)1 and 3

(B)1 and 2

(C)2 and 3

(D)1, 2 and 3

Answer Key

Q.1 Ans. [A]

Q.2 Ans. [D]

Q.3 Ans. [D}

Q.4 Ans. [C]

Q.5 Ans. [0, Range -0.01 to 0.01]

Q.6 Ans. [B]

Q.7 Ans. [C]

Q.8 Ans. [A]

Q.9 Ans. [8]

Q.10 Ans. [0]

Q.11 Ans. [56.54]

Q.12 Ans. [C]

Q.13 Ans. [B]

Q.14 Ans. [D]

Q.15 Ans. [1.062]

Q.16 Ans. [78.78]

Q.17 Ans. [B]

Q.18 Ans. [6]

Q.19 Ans. [A]

Q.20 Ans. [B]

Q.21 Ans. [B]

Q.22 Ans. [C, D]

Q.23 Ans. [C]

Q.24 Ans. [6.43]

Q.25 Ans. [D]

Q.26 Ans. [A]

Q.27 Ans. [B]

Q.28 Ans. [B]

Q.29 Ans. [A]

Q.30 Ans. [A]

Q.31 Ans. [D]

Q.32 Ans. [C]

Q.33 Ans. [A]

Q.34 Ans. [-1438]

Q.35 Ans. [46.91]

Q.36 Ans. [37.33]

Q.37 Ans. [B]

Q.38 Ans. [D]

Q.39 Ans. [D]

Q.40 Ans. [A]

Q.41 Ans. [A]

Q.42 Ans. [A]

2 Magnetostatics

MCQ, MSQ and NAT

Q.1 If the electric and magnetic field intensities in a medium are $3\sin(t-5z)\hat{a}_x$ V/m and $4\cos(t-5z)\hat{a}_y$ A/m respectively, then at z = 0, the instantaneous power density vector is-

(A)$12\sin t\cos t\hat{a}_z W/m^2$

(B)$3\sin(t-5)\cos(t-5)\hat{a}_z W/m^2$

(C)$4\sin t\cos t\hat{a}_z W/m^2$

(D)$12\cos 2t\hat{a}_z W/m^2$

Q.2 Consider a wire loop having an area of 0.16 m^2. It is placed in a uniform magnetic field of strength 0.2 T in such a way that the plane of the loop is perpendicular to the field. After two seconds, the magnetic field reverses its direction. The magnitude of the average emf induced in the loop during this time is_____V.

(Up to 2 decimal places)

Q.3 The plane of a 500-turn square coil of side length 40 cm makes an angle of 60° with a uniform magnetic field. The magnitude of the magnetic field changes by 8 T in 1 second. If the direction of the magnetic field is fixed, then the magnitude of the induced EMF is _____(volts) in the coil. (Rounded off to the nearest integer)

Q.4 A circular coil with 120 turns, 10 cm radius, and resistance of $10\ \Omega$ is placed in a uniform magnetic field in such a way that it is perpendicular to the plane of the coil. The field changes uniformly from zero to 0.4 T in 5 seconds. The induced current established in the coil during this change of field is_____ mA.

Q.5 Which of the following statement(s) is/are correct?

(A)$\nabla\times\vec{H}=\vec{J}$ is known as Ampere's circuit law in point form or differential form.

(B)$\nabla\times\vec{H}=\vec{J}$ is also known as Maxwell's third equation.

(C)The magnetostatic field is conservative as $\nabla\times\overrightarrow{H}=\overrightarrow{J}\neq 0$

(D)Ampere's circuit law can be applied irrespective of the shape of the closed curve 'C'.

Q.6 Consider an electric power line that carries 1500 A of current. It is located at a place where it experiences 5×10^{-5} T of the magnetic field. This power line makes 75° angle with respect to the magnetic field. Determine the magnitude of the magnetic force on the line if it is 100 m long.

(A)9.11 N (B)14.48 N

(C)7.24 N (D)3.62 N

Q.7 In a conducting medium,

$\vec{H} = y^2 z\hat{a}_x + 2(x+1)yz\hat{a}_y - (x+1)z^2\hat{a}_z \text{ A/m}$. The conduction current density at $(1, 0, -3)$ is-

A) $9\hat{a}_x \text{ A/m}^2$

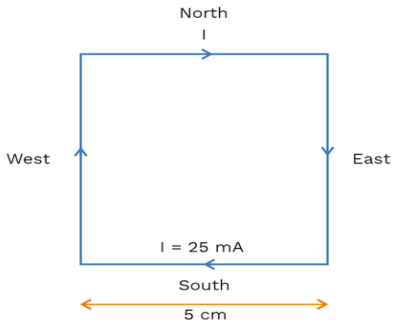
(B) $9\hat{a}_y \text{ A/m}^2$

(C) $4.5\hat{a}_x \text{ A/m}^2$

(D) $4.5\hat{a}_y \text{ A/m}^2$

Q.8 If a wire loop is bent into the shape of a square with each side of 5 cm. This loop is placed horizontally on a tabletop in such a way that its two sides are oriented north/south and the other two sides are oriented east/west.

A battery is also connected so that a current of 25 mA is produced around the loop and this current flows in the clockwise direction as shown below.



Calculate the overall torque magnitude on the loop, if it experiences a magnetic field of 0.5 × 10^{-4} T.

(A)6.245 N-m

(B)5.625 N-m

(C)12.245 N-m

(D)3.125 nN-m

Q.9 Consider a radial field $\overrightarrow{\mathrm{H}} = \frac{2.5 \times 10^6}{\mathrm{r}} \cos\phi \hat{\mathrm{a}}_\mathrm{r} \mathrm{A/m}$ existing in free space.

The magnetic flux crossing the surface defined by $0 \le \phi \le \frac{\pi}{4}$ and $0 \le \mathrm{z} \le 1 \text{ m}$ will be _____Wb. (Rounded off up to two decimal places) $\left[\mu_0 = 4\pi \times 10^{-7} \mathrm{H/m}\right]$

Q.10 Find the magnetic flux density vector $\vec{B}$ at point P(2, 6, 4) due to a current element of $2\pi\,(0.3\hat{a}_x - 0.4\hat{a}_y)\,A$ located point Q (8, -4, 6).

[Consider $\mu_0 = 4\pi \times 10^{-7}$ H/m.]

(A) $\frac{\pi}{140\sqrt{140}}[0.8\hat{a}_x + 0.6\hat{a}_y + 0.6\hat{a}_z] \times 10^{-7}$ Tesla

(B) $\frac{2\pi}{140\sqrt{140}}[0.8\hat{a}_x - 0.6\hat{a}_y + 0.6\hat{a}_z] \times 10^{-7}$ Tesla

(C) $\frac{2\pi}{140\sqrt{140}}[0.8\hat{a}_x + 0.6\hat{a}_y + 0.6\hat{a}_z] \times 10^{-7}$ Tesla

(D) $\frac{\pi}{140\sqrt{140}}[0.8\hat{a}_x - 0.6\hat{a}_y + 0.6\hat{a}_z] \times 10^{-7}$ Tesla

Q.11 Consider the following statements regarding Maxwell's equations for time-varying fields. Which of the following statement(s) is/are correct?

(A) $\vec{\nabla} \cdot \vec{D} = \rho_v$

(B) $\vec{\nabla} \cdot \overrightarrow{\mathrm{B}} = \overrightarrow{\mathrm{k}}$

(C) $\vec{\nabla} \cdot \vec{E} = \frac{\partial \vec{B}}{\partial \mathrm{t}}$

(D) $\vec{\nabla} \times \overrightarrow{\mathrm{H}} = \sigma \overrightarrow{\mathrm{E}} + \varepsilon \frac{\partial \overrightarrow{\mathrm{E}}}{\partial \mathrm{t}}$

Q.12 A straight conductor of length 0.65 m carries a current of 2.5 A in the $-\hat{\mathrm{a}}_\mathrm{z}$ direction. If the magnetic field is given as $\vec{B} = 3.50 \times 10^{-3}\,(\hat{a}_x - \hat{a}_y)\,\mathrm{T}$, then the force on the conductor is-

(A)$5.687 \times 10^{-3}\,(-\hat{a}_x - \hat{a}_y)\,N$
(B)$7.42 \times 10^{-3}\,(-\hat{a}_x + \hat{a}_y)\,N$

(C)$7.42 \times 10^{-3}\,(\hat{a}_x - \hat{a}_y)\,N$
(D)$5.687 \times 10^{-3}\,(\hat{a}_x + \hat{a}_y)\,N$

Q.13 A conductor carries a current of 20A and is at right-angles to a magnetic field having a flux density of 0.9T. If the length of the conductor in the field is 30cm. Calculate the force acting on the conductor.

(A)2.7N
(B)5.4N

(C)54N
(D)27N

Q.14 The force on the current carrying conductor in a magnetic field depend upon:

1. The flux density of the field

2. The strength of the current

3. The length of the conductor perpendicular to the magnetic field

4. The directions of the field and the current

(A) 1, 3 and 4 only
(B)1, 2 and 3 only

(C) 1, 2 and 4 only
(D)1, 2, 3 and 4

Q.15 "The line integral of magnetic field intensity H around a closed path is exactly equal to the direct current enclosed by that path" is the statement of -

(A)Divergence theorem
(B)Ampere's Circuital Law
(C)Biot-Savart Law
(D)Coulomb's Law

Q.16 What is the flux density of a magnet having cross-section of 43 cm² and exerting a flux of 2.7 mWb.

(A)0.54 T
(B)0.63 T

(C)1.85 T
(D)1.59 T

Q.17 Maxwell's 3rd equation is derived from -

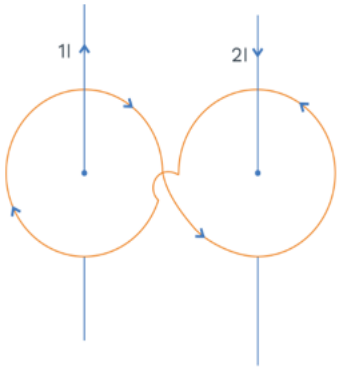
(A)Ampere's law

(B)Faraday law of electromagnetic induction

(C)Non-existence of isolated magnetic pole

(D)None of these

Q.18 Consider the given figure:



What is the circulation of $\vec{H}$ in close contour C.

(A)3I (B)–3I

(C)I (D)–I

Q.19 Match list-X (laws) with list-Y (Applications) and select the correct answer

List-X	List-Y To find the:
I. Coulomb's law	1. Force on a charge
II. Ampere's law	2. Electric flux density at a point
III. Biot-savart's law	3. Magnetic flux density at a point
IV. Gauss's law	4. Force due to current carrying conductors

(A)I-1, II-3, III-4, IV-2
(B)I-1, II-3, III-2, IV-4

(C)I-2, II-3, III-4, IV-1
(D)I-2, II-3, III-1, IV-4

Q.20 Assertion **(A)**: The solution of Poisson's equation is the same as the solution of Laplace's equation.

Reason **(R)**: Laplace equation is a special case of Poisson's equation of source-free region.

(A)Both Assertion(A) and Reason(R) are true and Reason(R) is the correct explanation of Assertion(A).

(B)Both Assertion(A) and Reason(R) are true but Reason(R) is not the correct explanation of Assertion(A).

(C)Assertion(A) is true but Reason(R)

is false.

(D)Assertion(A) is false but Reason(R) is true.

Q.21 A single turn circular coil having diameter of 50 m carries a current of 25×10^4 A. The relative permeability of the space surrounding the coil is unity. Find the magnetizing force at a point on the axis of the coil and 80 m from the coil. Assuming the Laplace's expression for magnetizing force due to a current element.

(A)98.91 AT/m
(B)115.25 AT/m

(C)132.58 AT/m
(D)152.14 AT/m

Q.22 The magnetic flux through each turn of coil consisting of 220 turns is $\left(t^2 - 6t\right)\, mWb$, where t is the time in seconds. The induced emf in the coil at t = 5 sec is -

(A)–0.88 V
(B)–1 V

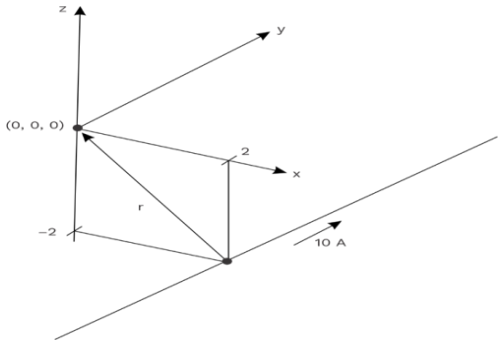
(C)0.88 V
(D)8.8 V

Q.23 A flux of 2.3 mWb is exerted by a magnet having a cross-section of 60 cm². The flux density in tesla is

(A)0.38×10^{-4} T
(B)3.8 T

(C)0.38×10^{-3} T
(D)0.38 T

Q.24 A current filament of 10 A in the $\hat{a}_y$ direction is parallel to the y-axis at x = 2 m and z = -2m, as shown in the figure below. The magnetic field intensity $(\vec{H})$ at the origin is-



(A) $0.562\left(\frac{\hat{a}_x + \hat{a}_y}{\sqrt{2}}\right)$ A/m

(B) $-0.281\left(\frac{\hat{a}_x + \hat{a}_y}{\sqrt{2}}\right)$ A/m

(C) $-0.562\left(\frac{\hat{a}_x+\hat{a}_y}{2}\right)\mathrm{A/m}$

(D) $0.281\left(\frac{\hat{a}_x+\hat{a}_y}{\sqrt{2}}\right)\mathrm{A/m}$

Q.25 A square loop of side 2 m is placed with its center at the origin in the x-y plane in a region where the magnetic flux density is given as

$\vec{B}=(2y\hat{a}_x-3z\hat{a}_y+4\hat{a}_z)\,e^{-2t}\ \mathrm{Wb/m^2}$

The induced EMF (in volts) at t = 2 sec in the loop will be-

(A)1747.14
(B)0.658

(C)58.6
(D)0.586

Q.26 In a certain conducting region, $\mathrm{H}=-3\mathrm{y}^2\hat{\mathrm{a}}_\mathrm{x}\mathrm{A/m}$. The current through a square of side 1 m with one corner at origin and sides coinciding with positive x and y axis is-

(A)1.5A

(B)3A

(C)6A
(D)12A

Q.27 Consider a circular loop conductor lying in the z = 0 plane, that has a 0.1 m radius and 4Ω resistance.

The flux density is given as –

$\vec{B}=0.5\sin 10^3\mathrm{t}\hat{a}_z(\,\mathrm{Tesla}\,)$

The current in the loop is –

(A)$0.5\pi\cos 10^3t\,(\hat{a}_x)$ Amperes
(B)$1.25\pi\cos 10^3t\,(-\hat{a}_y)$ Amperes
(C)$1.25\pi\cos 10^3t\,(-\hat{a}_z)$ Amperes
(D)$5\pi\cos 10^3t\,(-\hat{a}_y)$ Amperes

Answer Key

Q.1 Ans. [A]

Q.2 Ans. 0.32]

Q.3 Ans. [320]

Q.4 Ans.30.1

Q.5 Ans. [A, B, D]

Q.6 Ans. [C]

Q.7 Ans. [B]

Q.8 Ans. [D]

Q.9 Ans. [2.22]

Q.10 Ans. [C]

Q.11 Ans. [A, D]

Q.12 Ans. [A]

Q.13 Ans. [B]

Q.14 Ans. [D]

Q.15 Ans. [B]

Q.16 Ans. [B]

Q.17 Ans. [B]

Q.18 Ans. [B]

Q.19 Ans. [A]

Q.20 Ans. [D]

Q.21 Ans. [C]

Q.22 Ans. [C]

Q.23 Ans. [D]

Q.24 Ans. [A]

Q.25 Ans. [D]

Q.26 Ans. [B]

Q.27 Ans. [B]

3 Plane Wave Propagation

MCQ, MSQ and NAT

Q.1 Calculate the angle of refraction if a wave traveling in the air enters the water at 16° with the normal. (Assume the refractive index of water as 1.33)

(A)22° (B)33.13°

(C)11.96° (D)40.21°

Q.2 The Brewster's angle of a wave traveling into flint glass from water is _____(in degrees and rounded off up to two decimal places).

[The refractive index of the given glass is 1.49 and that of water is 1.33]

Q.3 An electromagnetic wave is propagating in the z-direction in a lossy medium. The attenuation constant is 0.55 Np/m and the amplitude of the electric field of the wave is 110 V/m at z = 0. How far will the wave travel before the amplitude of the EM wave reduce to 10 V/m?

(A)4.35 cm (B)0.435 m

(C)2.17 m (D)4.35 m

Q.4 If a wave is incident from air to the glass at Brewster's angle, then the transmitted angle is-

[Assume that the refractive index of glass is 1.45].

(A)69.18° (B)17.29°

(C)34.6° (D)5.88°

Q.5 Consider a plane wave propagating at a 2 MHz frequency in fresh water. Considering the losses in water at the given frequency are negligible, the phase velocity and the wavelength in water are respectively-

[Assume $\mu_r = 1$ and $\varepsilon_r = 81$].

(A)16.53×10^6 m/s and 3.3 cm
(B)3.3×10^7 m/s and 16.53 m

(C)16.53 m/s and 3.3×10^7 m
(D)3.3×10^6 cm/s and 16.53 cm

Q.6 If $\vec{E}(z,t) = 2000\cos\left(2\pi 10^7 t - \beta z\right)\hat{a}_x V/m$ and $\vec{H}(z,t) = 5.8\cos\left(2\pi 10^7 t - \beta z\right)\hat{a}_y A/m$ are the electric and magnetic fields of a uniform plane wave propagating at a velocity of 2.4×10^8 m/s in a perfect dielectric lossless medium, then the relative permeability (μ_r) is _____ (Rounded off up to two decimal places)

$[Assume\ \varepsilon_r = 1]$

Q.7 A conductor has relative permittivity, $\varepsilon_r = 2.4$ and conductivity, $\sigma = 4.5 \times 10^{-4}\ S/m$. The skin depth for a frequency of 25 kHz is _____meters.(Rounded off to the nearest integer).

[Assume $\varepsilon_0 = \left(\frac{10^{-9}}{36\pi}\right) F/m, \mu_r = 1$ and $\mu_0 = 4\pi \times 10^{-7} H/m$

Q.8 If the skin depth at 16 MHz for a good conductor is 0.04 m, then the phase velocity of an electromagnetic wave at 1600 MHz in the same conductor is-

(A)10.21 × 10^6 m/s
(B)20 × 10^6 m/s

(C)4.21 × 10^6 m/s
(D)40.21 × 10^6 m/s

Q.9 Evaluate the surface resistance of the silver coating used on a glass material for making it a good conductor at a frequency of 6000 MHz. The conductivity of silver is σ = 8.1 × 10^6 S/m, and μ_r = 1.

[Consider $\mu_0 = 4\pi \times 10^{-7} \mathrm{H/m}$]

(A)54 mΩ
(B)0.027 mΩ

(C)27 Ω
(D)0.54 mΩ

Q.10 A sinusoidal electric field with an amplitude of 200 V/m and frequency of 1.5 GHz exists in a lossy dielectric medium. This medium has ε_r = 2.5 and a loss tangent of 0.004. The conductivity in mS/m is_____(Up to two decimal places).

[Assume $\varepsilon_0 = \left(\frac{10^{-9}}{36\pi}\right) \mathrm{F/m}$]

Q.11 Evaluate the average power crossing a circular area with a 500 cm radius present in a plane where z is constant. The electric field of the electromagnetic wave in the free space is $\vec{E} = 60\cos(\omega t - \beta z)\hat{a}_x$ V/m.

(A)300 W
(B)150 W

(C)500 W
(D)375 W

Q.12 An EM wave is propagating through Teflon whose length is 200 cm. If the frequency is 12 GHz and the loss tangent at 25°C is 4 × 10^{-4}, then the attenuation constant(in Np/m) is _____. [Up to three decimal places]

[Given, $\varepsilon_r = 2.08$, μ_r = 1 and $\varepsilon_0 = \left(\frac{10^{-9}}{36\pi}\right)$ F/m]

Q.13 Consider a uniform plane wave that is incident from the air on the glass at an angle of 30° to the normal. For P-polarization, the fraction of the incident power that is reflected is_____ (Rounded off up to two decimal places)

[Assume that the glass has a refraction index of 1.45, the air has a refractive index of 1, and the intrinsic impedance of air is 377 Ω]

Q.14 Evaluate the effective impedance for s-polarization if a uniform plane wave is incident from air on the glass at an angle 45° with the normal.

[Assume η_{air} = 377 Ω]

(A) 600 Ω (B) 533.15 Ω

(C) 233.30 Ω (D) 250 Ω

Q.15 Consider the region-1 with z < 0 and region -2 with z > 0 and both are dielectrics with μ = μ_0. A plane uniform wave travels in $\hat{a}_z$ direction having a frequency of 8 × 10^{10} rad/s. The wavelength of region-1 and region-2 is 0.08 m and 0.06 m respectively.

Which of the following statement(s) is/are correct?

(A)The fraction of incident power that is reflected is 2.04 × 10^{-2}

(B)The Standing Wave Ratio is 1.33 in region-1.

(C)The fraction of the incident power that is transmitted is 20.04

(D)The reflection coefficient is $\left(-\frac{1}{9}\right)$

Q.16 Which of the following statement(s) is/are correct for a plane EM wave travelling in a vacuum along the z-direction?

(A)The electric and magnetic fields are in the x-y plane.

(B)The wavelength of an EM wave is 10 m, if its frequency is 30 MHz.

(C)The electric and magnetic fields are in the z-direction.

(D)The wavelength of an EM wave is 20 m, if its frequency is 30 GHz.

Q.17 If the amplitude of the magnetic field of a harmonic electromagnetic wave in a vacuum is 1000 nT, then the amplitude of the electric field of the wave is _____N/C. [Rounded off to the nearest integer]

Q.18 Consider a uniform plane wave that is incident from the air on the glass at an angle of 30° to the normal. For P-polarization, the fraction of incident power that is reflected is _____(Rounded off up to two decimal places)

[Assume that the glass has a refraction index of 1.45, the air has a refractive index of 1, and the intrinsic impedance of air is 377 Ω]

Q.19 Find the skin depth at $10\mathrm{GHz}$ for silver with conductivity of $6.1 \times 10^7\ \mathrm{S/m}$ and relative permittivity 1.

(A)6.443 $\mu\mathrm{m}$ (B)64.43$\mu\mathrm{m}$

(C)644.3$\mu\mathrm{m}$ (D)0.6443$\mu\mathrm{m}$

Q.20 A uniform plane wave in air partially reflects from the surface of a material whose properties are unknown. The measurements of the electric field in the region in front of the interface field yield a 2 m spacing between maximas& the first maxima occurs at 1 m from the interface. The VSWR was measured as 4. Then, the intrinsic impedance of the unknown material is _____ Ω. [Rounded off to the nearest integer]

Q.21 A uniform plane wave of frequency 5 GHz is incident normally on a dielectric plate having a thickness of 5 mm. If the phase shift during transmission through the plate is 60°, then the dielectric constant of the plate is _____. (Rounded off to the nearest integer).

Q.22 A plane wave is having magnetic field vector as

$$\vec{H} = \frac{1}{2}e^{-0.2x}\cos\left(10^7 t - 1.5x\right)\hat{a}_z \mathrm{A/m}.$$

Which of the following statement(s) is/are correct?

(A)The wavelength of the wave is 4.189 m

(B)The propagation constant of the wave is (-0.2 – j1.5) m^{-1}

(C)The wave travels along the +x direction.

(D)The wave travels along the +z direction.

Q.23 An EM wave of frequency f = 0.5 MHz is traveling through a medium of

$$\sigma = 0, \varepsilon_r = 6 \text{ and } \mu_r = 7.$$

Which of the following statement(s) is/are correct?

(A)Propagation constant, γ= j6.78 × 10^{-2} m^{-1}
(B)Propagation constant, γ = j6.78 m^{-1}

(C)Intrinsic Impedance, $\eta = 407\ \Omega$
(D)Intrinsic Impedance, $\eta = 20\ \Omega$

Q.24 A good conductor is having conductivity $\sigma = 3.2 \times 10^7 \mho/\mathrm{m}$ at frequency, f = 10 MHz and $\mu_r = 1$. The skin depth is _____μm. (Up to 2 decimal places)

Q.25 A poor conductor is having conductivity $\sigma = 1 \times 10^{-3}\mathrm{mho/m}$ and permittivity $\varepsilon = 30\varepsilon_0$. The relaxation time of the conductor is _____ x 10^{-9} sec. (Up to 1 decimal place)

Q.26 If a good conductor has conductivity 6.6 × 10^7 S/m and relative permittivity of 1, then the skin depth at a frequency of 11 GHz is-

(A)1.96 μm
(B)3.83 × 10^{-14} m

(C)0.196 μm
(D)3.83 pm

Q.27 The intrinsic impedance 'η_0' for free space is given by-

(A)$\sqrt{\frac{\mu_0}{\varepsilon_0}}$ (B)$\frac{\mu_0}{\varepsilon_0}$

(C)$\sqrt{\mu_0\varepsilon_0}$ (D)$\mu_0\varepsilon_0$

Q.28 A uniform plane wave is propagating in the air having the electric field component as $\vec{E} = 15e^{j(4x+8z)}\hat{a}_y V/m$.

The frequency of the operating wave is _____ GHz (up to 3 decimal places).

Q.29 The velocity of light in certain non-magnetic medium is $\frac{1}{3}$ times of velocity of light in air. What is the relative permittivity of the medium?

(A)1.73 (B)3

(C)9 (D)1

Q.30 The skin depth at 3 MHz is 5 cm. Calculate the skin depth at 18 MHz.

(A)1.8 cm (B)1.5 cm

(C)2.5 cm (D)2.04 cm

Q.31 The ratio of magnitude of conduction current to displacement current is given by

(A)Conduction ratio
(B)Dissipation factor

(C)Loss tangent

(D)Intrinsic ratio

Q.32 An electromagnetic wave is transmitted into a conducting medium of conductivity 'σ'.

Which one of the given statements is true regarding the depth of penetration?

(A)It is directly proportional to frequency.

(B)It is inversely proportional to frequency.

(C)It is inversely proportional to the square root of frequency.

(D)It is directly proportional to the square root of frequency.

Q.33 Polarization is a measure of-

(A)Dielectric constant per unit volume
(B)Voltage gradient to produce electric breakdown

(C)Product of charge and distance.
(D)Excess charge density.

Q.34 Which of the following statements are correct?

1. The net power flowing out of a given volume is equivalent to the time rate decrease in the energy stored with the volume minus ohmic losses.

2. The integration of the pointing vector over a closed surface will provide the total power flowing.

3. The Poynting vector is parallel to both $\vec{E}$ and $\vec{H}$.

4. The Poynting vector is time-invariant.

(A)1, 2, and 3. (B)2, 3, and 4.

(C)1 and 2. (D)1, 2, and 4.

Q.35 Consider the following statements regarding wave polarization:

1. A uniform plane wave is considered to be linearly polarized when it has one component and its transverse components are in the same phase.

2. The locus of the tip of the electric field is a function of time is regarded as the polarization.

3. The circular polarization occurs when the magnitude of x and y components are the same but the phase difference between them is even multiple of π.

4. Circular polarization is a special case of elliptical polarization.

Which of the above statements are correct?

(A)1 and 2 only. (B)1 and 3 only.

(C)1 and 4 only. (D)1, 2 and 4.

Q.36 The angle of transmission of a wave, incident at an angle of 30° from air to Teflon

$(\epsilon_r = 2.1)$

(A)$\theta_t = 35.53°$

(B)$\theta_t = 42.17°$

(C)$\theta_t = 20.18°$

(D)$\theta_t = 46.43°$

Q.37 An EM wave travels in free space with the electric field component-

$$E_s = 100e^{j(0.5y+0.866z)}\hat{a}_x$$

What will be the value of wavelength 'λ'?

(A)5.462 m

(B)6.283 m

(C)8.283 m

(D)3.462 m

Q.38 The correct statements about skin depth are-

1. Skin depth is a measure of the depth to which an EM wave can penetrate the medium.

2. Skin depth is given as-

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} = \frac{1}{\alpha}$$

3. The distance δ through which the wave amplitude decreases to a factor e^{-1} (about 37% of the original value) is called skin depth or penetration depth of the medium.

i.e.$E_0 e^{-\alpha\delta} = E_0 e^{-1}$

or

$$\delta = \frac{1}{\alpha}$$

(A)1 and 2 only
(B)1, 2 and 3

(C)2 and 3 only
(D)None

Q.39 The conduction current density in a lossy dielectric medium is $\overrightarrow{J_c} = 0.02 \sin 10^9 t \ \mathrm{A/m^2}$. The displacement current density, if σ = 10^3 S/m and ε_r = 5.5, is-

(A)$0.973 \cos 10^9 \mathrm{t} \mu \mathrm{A/m^2}$

(B)$0.973 \sin 10^9 \mathrm{t} \mu \mathrm{A/m^2}$

(C)$1.51 \sin 10^9 \mathrm{t} \mu \mathrm{A/m^2}$

(D)$1.51 \cos 10^9 \mathrm{t} \mu \mathrm{A/m^2}$

Q.40 A copper conductor is having σ = 5.8 × 10^7 S/m and $\varepsilon_r = \mu_r$ = 1. If it supports a uniform plane wave at a frequency of 50 Hz, then the propagation constant is-

(A)$114 + j114 m^{-1}$

(B)$14.29 + \mathrm{j}14.29 \ \mathrm{m}^{-1}$

(C)$107 + j107 \ \mathrm{m}^{-1}$

(D)$107 + j114 m^{-1}$

Q.41 A plane wave traveling in air is incident normally on a material with ε_r= 3 and μ_r =1. The correct statement(s) is/are-

(A)The reflection coefficient is -0.268
(B)The transmission coefficient is 0.666

(C)The reflection coefficient is – 0.333
(D)The transmission coefficient is 0.731

Q.42 .A polarized wave is incident from air to a non-magnetic medium having ε_r = 6 and μ_r = 1 at the Brewster angle. The angle of incidence is _____ degrees. (Upto 2 decimal places).

Q.43 A parallel plate air-filled capacitor has plate area of 10^{-6} m^2 and plate separation of 1 mm. The capacitor is connected to 3/4 V, 2.4 GHz source. The displacement current is _____mA. (Up to one decimal place).

[Consider $\varepsilon_0 = \frac{10^{-9}}{36\pi} \ \mathrm{F/m}$]

Q.44 A TEM wave has power density of $1.4 \ \mathrm{W/m^2}$ in a medium with $\mu_r = 1 \ and \ \varepsilon_r = 4$. The maximum value of electric field intensity is _____V/m.

Q.45 The electric and magnetic fields in a coaxial cable are $E = \frac{10}{\rho \ln\left(\frac{b}{a}\right)} \cos \omega \left(t - z\sqrt{\mu_0 \varepsilon_0}\right) \hat{a}_\rho$ and $H = \frac{5}{2\pi\rho} \cos \omega \left(t - z\sqrt{\mu_0 \varepsilon_0}\right) \hat{a}_\phi$ respectively, where,

b = radius of the outer conductor

a = radius of the inner conductor

The time average power (in Watts) through the co-axial cable is-

(A)12.5 (B)25

(C)50 (D)75

Q.46 Consider the uniform plane wave in sea water having $\sigma = 8 \ \mathrm{s/m}, \quad \varepsilon = 75\varepsilon_0$ and $\mu = \mu_0$. The value of attenuation constant, α (in Np/m) at frequency of 20 kHz, is _____. (Correct up to three decimal places)

Q.47 A plane wave traveling is +y direction in a lossy medium $(\varepsilon_r = 4, \mu_r = 1, \sigma = 10^{-2} \ \mathrm{J/m})$ has $\mathrm{E} = 30 \cos\left(10^9 \pi \mathrm{t} + \frac{\pi}{4}\right) \mathrm{a_z V/m}$ at $\mathrm{y} = 0$. The distance travelled (in mm) by the wave to have a phase shift of 10° is _____ (Answer up to 3 decimals)

Q.48 An electromagnetic wave is propagating in the z-direction in a lossy medium. The attenuation constant is 0.55 Np/m and the amplitude of the electric field of the wave is 110 V/m at z = 0. How far will the wave travel before the amplitude of the EM wave reduce to 10 V/m?

(A)4.35 cm (B)0.435 m

(C)2.17 m (D)4.35 m

Q.49 A lossy material has $\mu = 5\mu_0, \varepsilon = 2\varepsilon_0$. If the frequency is 5MHz and the phase constant is 10 rad/m, then the correct statement(s) is/are

(A)Loss tangent is 1823
(B)Conductivity is 0.506 S/m

(C)Loss tangent is 100
(D)Conductivity is 9.54 S/m

Q.50 The distance (in meters) a wave has to propagate in a medium having a skin depth of 0.2 m so that the amplitude of the wave attenuates by 25dB is _____. (up to 3 decimal places)

Q.51 A uniform plane wave traveling in air is normally incident on a dielectric medium. The refractive index of the dielectric medium is 1.48. The percentage of the incident power that is reflected from the air-dielectric interface is _____. (Up to 2 decimal places)

Q.52 If $\vec{E} = (2\hat{a}_x + 2j\hat{a}_y)\, e^{jkz - j\omega t}$ and $\vec{H} = \dfrac{k}{\omega\mu}(\hat{a}_y + j\hat{a}_x)\, e^{\text{jkz}-\text{j}\omega t}$, the time averaged pointing vector is-

(A)Null vector (B)$\dfrac{k}{\omega\mu}\hat{a}_z$

(C)$\dfrac{2k}{\omega\mu}\hat{a}_z$ (D)$\dfrac{4\text{k}}{\omega\mu}\hat{\text{a}}_z$

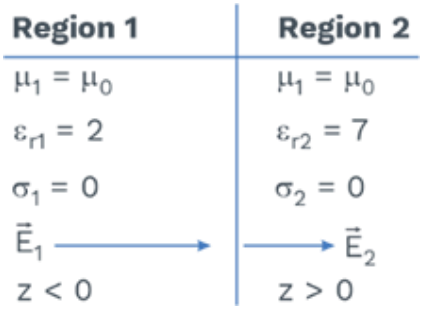
Q.53 If the given parameters are $\sigma = 3 \times 10^{-4}\ \text{S/m}$ and $\epsilon_r = 2.4$. The depth of the point at which the field strength for f = 30 kHz fall to 36.78% of that at the surface is

(A)168 m
(B)250 m

(C)150 m
(D)251 m

Q.54 Evaluate the displacement current density (in pA/m^2) in a current carrying conductor with 1 A of current flowing in it. Assuming a 50 Hz of frequency and wire is of copper having conductivity, σ = 5.0 × 10^7 (mho/m) and diameter 2 mm. Given ϵ_0 = 8.854 × 10^{-12} C/Nm^2 as in vacuum.

Q.55 Consider an electromagnetic wave with electric field $\vec{E}_1 = 5\vec{a}_x + 6\vec{a}_y + 7\vec{a}_z$ is incident normally on the interface from region 1 as shown below-



The electric field $\vec{\text{E}}_2$ in region 2 is–

(A)$\vec{\text{E}}_1 = \vec{\text{E}}_2$

(B) $\dfrac{10}{7}\vec{a}_x + 6\vec{a}_y + 7\vec{a}_z$

(C)$5\vec{a}_x + 6\vec{a}_y + 2\vec{a}_z$

(D) $\dfrac{10}{7}\vec{a}_x + \dfrac{12}{7}\vec{a}_y + 7\vec{a}_z$

Q.56 A plane wave has an electric field given by $\vec{\text{E}} = 10\text{e}^{-\text{z}/3}\sin\left(10^7\text{t} - \beta\text{z}\right)\vec{\text{a}}_\text{y}\ \text{V/m}$ w

hich is propagating through a medium having following parameters: $\varepsilon_r = 8.2 \; and \; \mu_r = 1.8 \; and \; \sigma = 0$. The wave impedance (in Ω) is _____.

(Up to 2 decimal places)

Q.57 Determine the average power (in W and up to two decimal places) crossing a circular area having a radius of 350 cm in the plane, where Z is constant.

The electric field in the free space is given as –

$$\overrightarrow{E}(z,t) = 45\cos(\omega t - \beta z)\hat{a}_x \mathrm{V/m}$$

Q.58 In a certain medium $(\varepsilon_r = 4, \mu_r = 4)$. A plane wave is propagating such that the electric field intensity of the wave is $\vec{E} = E_0 e^{-x/3} \sin\left(10^8 t - \beta x\right) \hat{a}_y \; \mathrm{V/m}$. The loss tangent of the medium will be _____.

Q.59 In free space complex electric field vector of a uniform plane wave is given by

$$\mathbf{E_s} = \left(\sqrt{3}\,\hat{\mathbf{a}_x} + \hat{\mathbf{a}_z}\right) \mathbf{e}^{-\mathrm{j}\frac{\pi}{25}(\sqrt{3}x - 2y - 3z)} \mathbf{V/m}$$

The apparent wavelengths along the x, y and z axes are $\lambda_x, \lambda_y, \lambda_z$.

Then $\lambda_x + \lambda_y + \lambda_z$ _____ (m)

Answer Key

Q.1 Ans. [C]

Q.2 Ans. [48.24]

Q.3 Ans. [D]

Q.4 Ans. [C]

Q.5 Ans. [B]

Q.6 Ans. [1.55]

Q.7 Ans. [150]

Q.8 Ans. [D]

Q.9 Ans. [A]

Q.10 Ans. [0.83]

Q.11 Ans. [D]

Q.12 Ans. [0.072]

Q.13 Ans. [0.021]

Q.14 Ans. [B]

Q.15 Ans. [A, B]

Q.16 Ans. [A, B]

Q.17 Ans. [300]

Q.18 Ans. [0.021]

Q.19 Ans. [D]

Q.20 Ans. [1508]

Q.21 Ans. [4]

Q.22 Ans. [A, C]

Q.23 Ans. [A, C]

Q.24 Ans. [28.13]

Q.25 Ans. [265.5]

Q.26 Ans. [C]

Q.27 Ans. [A]

Q.28 Ans. [0.425]

Q.29 Ans. [C]

Q.30 Ans. [D]

Q.31 Ans. [C]

Q.32 Ans. [C]

Q.33 Ans. [B]

Q.34 Ans. [D]

Q.35 Ans. [D]

Q.36 Ans. [C]

Q.37 Ans. [B]

Q.38 Ans. [B]

Q.39 Ans. [A]

Q.40 Ans. [C]

Q.41 Ans. [A, D]

Q.42 Ans. [80.53]

Q.43 Ans. [0.1]

Q.44 Ans. [22.97]

Q.45 Ans. [B]

Q.46 Ans. [0.794]

Q.47 Ans. [8.325]

Q.48 Ans. [D]

Q.49 Ans. [A, B]

Q.50 Ans. [0.576]

Q.51 Ans. [C]

Q.52 Ans. [A]

Q.53 Ans. [A]

Q.54 Ans. [17.7]

Q.55 Ans. [C]

Q.56 Ans. [176.63]

Q.57 Ans. [103.36]

Q.58 Ans. [0.52]

Q.59 Ans. [70.57]

4 Transmission Lines

MCQ, MSQ and NAT

Q.1 A 190 Ω lossless transmission line is terminated by a load impedance (200 + j110) Ω. If the wavelength is 75 cm, then the voltage standing wave ratio is _____. [Up to two decimal places]

Q.2 Consider a lossless transmission line of 100 cm in length and operating at a frequency of 140 MHz. The line parameters are L = 0.25 μH/m and C = 80 pF/m.

Which of the following statement(s) is/are correct?

(A)The phase velocity is 2.23×10^8 m/sec. (B)The characteristic impedance is 55.9 Ω.

(C)The phase constant is 3.94 rad/m. (D)None of the above

Q.3 A lossless long transmission line of length 3.5 m is air-filled. The line has a characteristic impedance of $120\pi\ \Omega$ and is terminated by a short circuit. The excitation frequency is 45 MHz. The input impedance for the transmission line (in ohms) is-

(A)j156.15 (B)j120 π

(C)j59.7 (D)–j120π

Q.4 The line parameters of a lossless transmission line are L = 0.36 μH/m and C = 120 pF/m. It operates at a frequency of 500 MHz and the length of the line is 100 cm.

The phase constant and the phase velocity respectively are-

(A)15.22 rad/m and 20.64×10^6 m/s
(B)20.64 rad/m and 1.52×10^8 m/s

(C)10.32 rad/m and 3.04×10^8 m/s
(D)20.64 rad/m and 6.59×10^6 m/s

Q.5 In a transmission line, determine the SWR if a load impedance of $70\angle 30°\Omega$ is connected to a 45 Ω characteristic impedance.

(A)1.032 (B)1.921

(C)2.048 (D)0.048

Q.6 The conductors of a two-wire transmission line have a radius of 0.5 mm and their center-to-center distance is 0.8 cm. The frequency is 150 MHz and $\sigma = 0, \sigma_c \to \infty$.

Which of the following statement(s) is/are correct?

(Consider $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m)

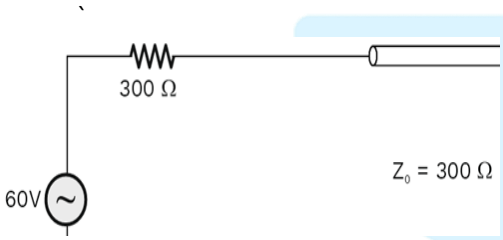
(A)The dielectric constant or relative permittivity of the insulating medium is $\epsilon_r = 1.23$ if $Z_0 = 300\Omega$

(B)The dielectric constant or relative permittivity of the insulating medium $\epsilon_r = 1.99$ if capacitance per unit length is C = 20 pF.

(C) The dielectric constant or relative permittivity of the insulating medium $\epsilon_r = 2.6$ if phase velocity is 2.6×10^8 m/s.

(D)The dielectric constant or relative permittivity of the insulating medium $\epsilon_r = 2.46$ if $Z_0 = 300\Omega$.

Q.7 Which of the following is/are correct for the lossless transmission line given below?



(A)The input impedance is 466 – j205.6 Ω

(B)The time average power dissipated in the generator internal impedance (Z_g)is 1.33 W

(C)The input impedance is 400 + j205.6 Ω

(D)The time average power dissipated in the generator internal impedance (Z_g) is 0.96 W

Q.8 A battery (50 V) having zero resistance is connected to a coaxial line which has a characteristic impedance of 60 Ω and the other end is terminated with an unknown resistive load. The voltage at the load end is 60 V. The load resistance or load impedance, and steady-state current through load resistance respectively are-

(A)150 Ω, 0.5 A (B)90 Ω, 0.55 A

(C)90 Ω, 0.2 A (D)150 Ω, 0.33 A

Q.9 A lossless transmission line is connected to a load composed of a 75 Ω resistor in series with a capacitor of capacitance C. The characteristic impedance is 50 Ω. If at 15 MHz frequency, the VSWR is 3, then the capacitance (C) of the capacitor is_____pF. [Up to two decimal places].



Q.10 Consider a 75 Ω lossless transmission line terminated by a purely resistive load and the VSWR is 3. The possible values of Z_L are-

(A)25 Ω (B)50 Ω

(C)75 Ω (D)225 Ω

Q.11 If the magnitude of the short-circuit input impedance and the characteristic impedance is 25 Ω and 50 Ω respectively, then the open-circuit input impedance is

(A)25 Ω (B)75 Ω

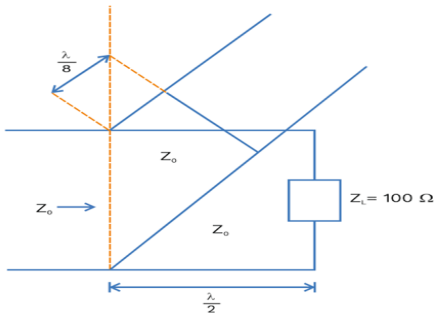
(C)100 Ω (D)50 Ω

Q.12 A transmission line is operating at 30 MHz with 10 µH/m inductance and 40 pF/m capacitance per unit length of the line. If the line is 50 cm in length,

then its electrical path length is _____π (radians).

[Up to one decimal place]

Q.13 A short-circuited stub is shunt connected to a transmission line as shown in the figure below.



If the characteristic impedance is 70 Ω, then the admittance Y seen at the junction of the stub and the transmission line is

(A)(0.014 – j0.01) ℧
(B)(0.01 – j0.014) ℧

(C)(0.014 + j0.01) ℧
(D)(0.01 + j0.014) ℧

Q.14 Calculate the length and the input impedance of a quarter-wave transformer that will match a 150 Ω load to a line with a characteristic impedance of 75 Ω at a frequency of 110 GHz.

(A)0.068 mm and 37.5 Ω
(B)0.68 m and 3.75 Ω

(C)0.0068 cm and 3.75 Ω
(D)0.068 cm and 37.5 Ω

Q.15 A rigid coaxial air line at 2 MHz has an inner and outer conductor diameter of 0.5 cm and 1 cm respectively. The conductors are made up of copper $\left(\mu = \mu_0 \; and \; \sigma_c \simeq 6 \times 10^7 \text{ S/m}\right)$.

The correct statement(s) is/are-

(A)Resistance per unit length is $3.45 \times 10^{-2} (\Omega/\text{m})$

(B)Inductance per unit length is $1.4\mu\text{H/m}$

(C)Conductance per unit length is 0 S/m

(D)Capacitance per unit length is 80.3 F/m

Q.16 Consider an unloaded underground cable at audio frequency. It has the following constants.

$$\text{R} = 40\Omega/\text{km}$$
$$\text{G} = 0.5\mu\text{mho/km}$$
$$\text{L} = 1\mu\text{H/km}$$
$$\text{C} = 0.08\mu\text{F/km}$$

Then at 400Hz,

(A)
$\text{Z}_0 = 446.03 \text{ m}\Omega; \alpha = 0.063 \text{ Np/km}, \beta = 0.063\text{rad/km}$

(B)
$Z_0 = 223.15\Omega, \alpha = 0.063 \text{ Np/km}, \beta = 0.036\text{rad/km}$

(C)
$\text{Z}_0 = 223.15\Omega, \alpha = 0.063 \text{ Np/km}, \beta = 0.063\text{rad/km}$

(D)
$\text{Z}_0 = 446.03\Omega, \alpha = 0.063 \text{ N}_\text{p}/\text{km}, \beta = 0.063\text{rad/km}$

Q.17 A transmission line with a characteristic impedance of 60 Ω is connected to a load that provides a reflection coefficient of 0.375. If the input to the line is 10 V, then the total power delivered to the load is_____mW. (Up to 1 decimal place)

Q.18 Which of the following statement(s) is/are correct for a lossless transmission line?

(A)$Z_{in} = -jZ_0$ for an open line with

l = $\lambda/2$.

(B)$Z_{in} = jZ_0$ for a shorted line with

l = $\lambda/8$.

(C)$Z_{in} = j\infty$ for a shorted line with

l = $\lambda/4$

(D)$Z_{in} = Z_0$ for a matched line of any length.

Q.19 Which of the following statement(s) is/are correct for a distortion-less transmission line?

(A)$\alpha = 0$
(B)$\beta = \omega\sqrt{LC}$

(C)$\alpha = \sqrt{RG}$
(D)$\frac{R}{L} = \frac{G}{C}$

Q.20 A 25Ω transmission line is terminated with an open-circuited load. The correct statement(s) is/are

(A) Reflection coefficient ($\overline{|L|}$) is 0

(B) VSWR is ∞

(C) Reflection coefficient ($\overline{|L|}$) is 1

(D) VSWR is 1

Q.21 The intrinsic impedance 'η_0' for free space is given by-

A)$\sqrt{\frac{\mu_0}{\varepsilon_0}}$ (B)$\frac{\mu_0}{\varepsilon_0}$

(C)$\sqrt{\mu_0\varepsilon_0}$ (D)$\mu_0\varepsilon_0$

Q.22 For a transmission line, if $Z_L = \frac{1}{2}jZ_0$, then the VSWR of the line is-

(A)∞ (B)2

(C)0.5 (D)1

Q.23 The reflection coefficient of a short-circuited line for voltage is-

(A)-1 (B)+1

(C)0.5 (D)Zero

Q.24 The range of VSWR and reflection coefficient respectively are -

(A)$-\infty$ to ∞ and 0 to 1
(B)0 to ∞ and -1 to 1

(C)1 to ∞ and -1 to 1
(D)0 to ∞ and 0 to 1

Q.25 When VSWR is 2, the magnitude of the reflection coefficient is____.

(A)$\frac{1}{4}$ (B)$\frac{1}{3}$

(C)$\frac{1}{2}$ (D)1

Q.26 The VSWR of the system in which voltmeter readings at maxima and minima are 64 mV and 32 mV respectively will be –

[Assume that the square-law detector is used to detect the signal level]

(A)0 (B)4

(C)1 (D)2

Q.27 The reflection coefficient for the line having characteristic impedance of $300\angle 0^\circ\Omega$ and load impedance is $150\angle 0^\circ\Omega$is -

(A)0.33 (B)1.33

(C)3 (D)3.33

Q.28 Consider a lossless line having characteristic impedance of 75 Ω and phase constant of 5 rad/m at frequency 150 MHz. The inductance per meter of the line is approximately -

(A)398 nH/m (B)350 nH/m

(C)70.73 nH/m (D)293 nH/m

Q.29 A transmission line (60 Ω) is terminated by a (100 + j80)Ω, the maximum impedance over the line is -

(A)100 Ω (B)200 Ω

(C)180 Ω (D)120 Ω

Q.30 Choose the incorrect statement regarding transmission lines.

(A)The characteristic impedance Z_0 of the line is the ratio of the positively traveling voltage wave to the current wave at any point on the line.

(B)A transmission line is said to be lossless if the conductors of the line are

perfect $(\sigma_c \approx \infty)$ and the dielectric medium separating them is lossless $(\sigma \approx 0)$

(C)A distortionless line is one in which the attenuation constant α is frequency independent while the phase constant β is linearly dependent on frequency.

(D)A distortionless line is one in which the attenuation constant α is linearly dependent on frequency and the phase constant β is frequency independent.

Q.31 A lossless transmission line has Z_0 = 100 Ω and it is operating at ω = 100 Mrad/sec. If the phase velocity on the line is 2.4×10^8 m/sec, then the distributed parameters L and C are respectively-

(A)0.50 µH/m & 35 pF/m
(B)0.35 µH/m & 50 pF/m

(C)41.6 µH/m & 4.166 pF/m
(D)0.416 µH/m & 41.66 pF/m

Q.32 A 40 Ω lossless line connects a signal to a load of 60 Ω. The VSWR is_____(up to 1 decimal place).

Q.33 The return loss (in dB) of transmission line having the voltage standing wave ratio (VSWR) equal to 7, is _____.

Q.34 Which of the following statement(s) is/are valid for perfectly matched load condition?

(A)The transmission line having a length less than $\frac{\lambda}{4}$ and short circuited at far end behaves as capacitance.

(B)The characteristics impedance Z_0 of a line is calculated when the length is equivalent to its wavelength.

(C)The standing wave ratio is 1.

(D)The transmission line of length $\left(\frac{\lambda}{4}\right)$ short-circuited at for end behaves as parallelly tuned circuit.

Q.35 A distortion-less transmission line has resistance of 0.08 Ω per unit length and characteristic impedance of 80 Ω . What will be the attenuation constant (in N_p /m)?

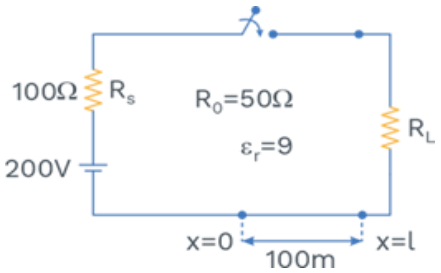
(Up to 3 decimal places)

Q.36 The minimum value of VSWR that may exist on a transmission line is

(A)1 (B)0

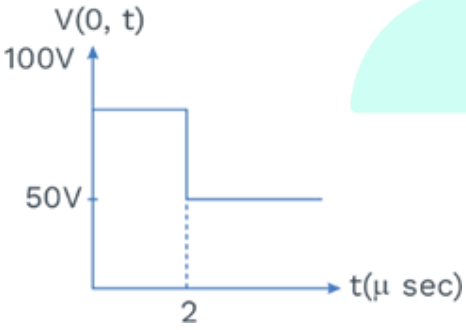
(C)10 (D)-1

Q.37 A 100 m long transmission line is shown in figure below.



It is terminated with a load R_L. The line is connected to a 200V source having source resistance R_s connected through a switch which closes at t = 0.

The transient response of the circuit at input of the line (x = 0) is given below-



Which of the following statement(s) is/are correct?

(A) $R_L = 22.73\Omega$
(B)$R_L = 27.22\Omega$

(C)Transient time = 1μsec

(D) Reflection coefficient at load, $\alpha_L = \frac{3}{8}$

Q.38 Consider a transmission line of 150 m length and a characteristic impedance of 600Ω, operating at 400 kHz with α = 2.4 × 10^{-3} Np/m. It supplies a load impedance of Z_R = 424.3 $\angle$ 45°Ω and β = 0.0212 rad m^{-1}.

The Z_S for the given transmission line at a length of 150 m is-

(A)600 $\angle$ 20.8° Ω
(B)116.6 $\angle$ 90° Ω

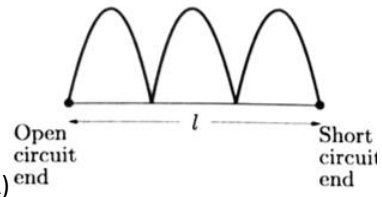
(C)502.7 $\angle$ 22.8° Ω
(D)–0.09 $\angle$ 0.20° Ω

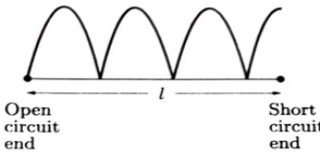
Q.39 Distance of the first voltage maximum and first current maximum from the load on 50Ω a lossless transmission line are respectively 4.5 cm and 1.5 cm. If the standing wave ratio on the transmission line is S = 3 then the load impedance connected to the transmission line will be

(A)$(90 - j120)\Omega$
(B)10Ω

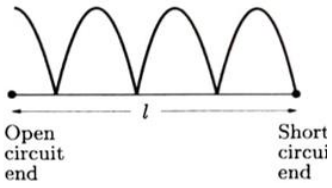
(C)$(30 - j40)\Omega$
(D)$(40 - j30)\Omega$

Q.40 A transmission line of length l is short circuited at one end open circuited at the other end. The incorrect voltage standing wave pattern in the transmission line will be

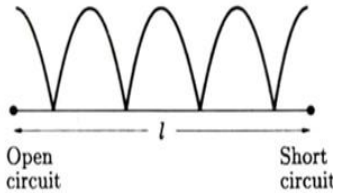
(A)


(B)


(C)

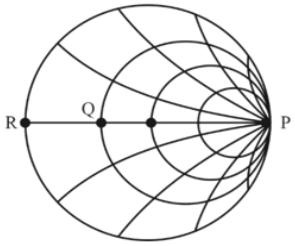


(D)



Q.41 At t = 0 a 50 Volt D. C. source with an internal resistance 30Ω is connected to a transmission line of 15Ω characteristic impedance having a load of 45Ω. The steady state load current for the transmission line is ____ Ampere.

Q.42



The points P, Q, and R are marked on the smith chart represents/shows

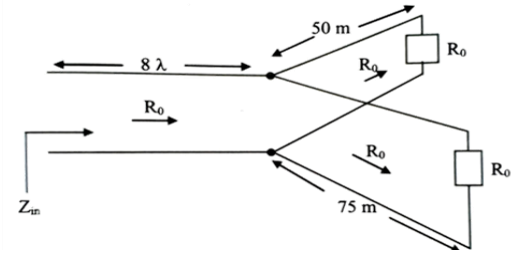
(A)R = Short circuit for impedance smith chart

(B)Q = (1, 0) means $Z_L = 1$

(C)P = Short circuit for susceptance smith chart

(D)Q = Z_L is infinity

Q.43 The input impedance of the transmission line configuration given below is _____. (Given $R_o = 25\Omega$)



Q.44 A 50Ω lossless transmission line of length $\frac{\lambda}{2}$ has been terminated with a normalized load impedance of (0.4 – j 0.3). The input impedance is Z_{in} of the line and magnitude of reflection coefficient is Γ. Choose the correct option/options.

(A)$Z_{in} = (20 - \text{j}15)\Omega$
(B)$Z_{in} = (20 + \text{j}15)\Omega$

(C)$\Gamma = 0.466$
(D)$\Gamma = 0.5$

Answer Key

Q.1 Ans. [1.75]

Q.2 Ans. [A, B, C]

Q.3 Ans. [C]

Q.4 Ans. [B]

Q.5 Ans. [C]

Q.6 Ans. [A, B]

Q.7 Ans. [A, D]

Q.8 Ans. [B]

Q.9 Ans. [160.4]

Q.10 Ans. [A, D]

Q.11 Ans. [C]

Q.12 Ans. [0.6]

Q.13 Ans. [B]

Q.14 Ans. [D]

Q.15 Ans. [A, C]

Q.16 Ans. [D]

Q.17 Ans. [295.9]

Q.18 Ans. [B, C, D]

Q.19 Ans. [B, C, D]

Q.20 Ans. [B, C]

Q.21 Ans. [A]

Q.22 Ans. [A]

Q.23 Ans. [A]

Q.24 Ans. [C]

Q.25 Ans. [B]

Q.26 Ans. [D]

Q.27 Ans. [A]

Q.28 Ans. [A]

Q.29 Ans. [C]

Q.30 Ans. [D]

Q.31 Ans. [D]

Q.32 Ans. [1.5]

Q.33 Ans. [2.498]

Q.34 Ans. [C, D]

Q.35 Ans. [0.001]

Q.36 Ans. [A]

Q.37 Ans. [A, C]

Q.38 Ans. [C]

Q.39 Ans. [C]

Q.40 Ans. [A, B, D]

Q.41 Ans. [0.67]

Q.42 Ans. [A, B]

Q.43 Ans. [12.5]

Q.44 Ans. [A, C]

5 Waveguides

MCQ, MSQ and NAT

Q.1 A rectangular waveguide operating in TE_{11} mode has dimensions 5 cm and 4 cm.

The cut-off frequency is _____ GHz. (Up to two decimal places).

Q.2 A circular waveguide with an inner diameter of 6.4 cm is operating in a dominant mode. The cut-off frequency is _____GHz. (Rounded off up to two decimal places).

Q.3 Consider a rectangular waveguide operating in TE_{mn} mode where n = 1 and the cut-off frequency is 7.5 GHz. The dimensions of the given waveguide are 4cm x 2cm. The value of 'm' is_____. (Rounded off up to nearest integer)

Q.4 A standard rectangular waveguide is operating in TM_{11} mode. The aspect ratio of 2:1 is considered, and the smaller dimension of the waveguide is 2 cm. The cut-off wavelength is-

(A)0.357 m (B)3.57 cm

(C)0.035 cm (D)3.57 m

Q.5 An air-filled rectangular waveguide is constructed to operate in TE_{10} mode having a cut-off frequency of 13.6 GHz. The same waveguide is now to be operated in TE_{01} mode with an operating frequency of 15 GHz. This operating frequency is 10% less than the cut-off frequency in TE_{01} mode. The required dimensions of the waveguide are-

(A)10 mm × 11 mm

(B)10 mm × 9 mm

(C)1.1 mm × 8.9 mm

(D)11 mm × 9 mm

Q.6 An S-band waveguide with the dimensions of 6.980 cm × 3.202 cm, is lossless, has air as the dielectric medium, and operates in the dominant mode. The wave propagates in the +z direction at 2.4 GHz. If the excitation level of the electric field is 10 kV/m, then the average transmitted power is _____ W. (up to 3 decimal places)

Q.7 For the dominant mode of operation in an air-filled circular waveguide with an inner diameter of 4 cm, the cutoff frequency is_____ GHz.(Up to three decimal places)

Q.8 The guided wavelength(in cm) for a rectangular waveguide having internal dimensions of 4.5 cm × 3 cm is_____.[Rounded off up to one decimal place]. Assume that a 12 GHz signal is propagating in the dominant mode.

Q.9 The cut-off wavelength λ_c for TE_{20} mode for a standard rectangular waveguide is-

(A)2/a (B)a

(C)2a (D)$2a^2$

Q.10 A rectangular waveguide is working in TE mode. If the cut-off frequency of the dominant mode is 9 GHz, then the inner dimension(a) of the wall is-

(A)1.67 cm (B)1.67 mm

(C)3.33 mm (D)3.33 cm

Q.11 An air-filled parallel-plate waveguide has the signal frequency of 2.5 GHz with the separation of 5 cm. For TM_2 mode, the cut-off frequency is-

(A)7.5 GHz
(B)4.5 GHz

(C)6 GHz
(D) 9 GHz

Q.12 A section of a rectangular waveguide of cross-section $2\ \text{cm} \times 1.5\ \text{cm}$ is to be used as a delay line in a radar at 10 GHz. What should be the length of the section for the delay of 10 nsec?

(A)19.8 m (B)1.98 m

(C)0.99 m (D)198 m

Q.13 Calculate the cut-off frequency for a waveguide having dimensions a = 3cm, $b = 1\ \text{cm}$, in TM_{11} mode.

(A)5 GHz
(B)31.6 GHz

(C)10 GHz
(D)15.8 GHz

Q.14 Assertion **(A)**: T_{EM} (Transfer Electro-Magnetic) waves cannot Propagate within a hollow waveguide of any shape.

Reason **(R)**: For a T_{EM} wave to exist within the waveguide, lines of the H field must be closed loops that require an axial component of E which is not present in a T_{EM} wave.

(A)Both Assertion**(A)** and Reason**(A)** are true and Reason**(R)** is the correct explanation of Assertion**(R).**

(B)Both Assertion**(A)** and Reason**(R)** are true but Reason**(R)** is not the correct explanation of Assertion**(A)**.

(C)Assertion**(A)** is true but Reason**(R)** is false.

(D)Assertion**(A)** is false but Reason**(R)** is true

Q.15 Match List I with List II and the correct answer using the codes given below the lists-

List-I(Transmission System)	List-II(Mode)
A. Rectangular waveguide	1. T_E/T_M
B. Circular Waveguide	2. T_{EM}
C. Coaxial line	3. Quasi-T_{EM}
D. Micro strip Line	

(A)A-1 B-1 C-2 D-3
(B)A-1 B-2 C-2 D-3

(C)A-2 B-2 C-3 D-3
(D)A-3 B-1 C-2 D-2

Q.16 The cut-off frequency for a circular waveguide having 4 cm of inner diameter and operating in a dominant mode will be -

(A)4.3 GHz (B)2.1 GHz

(C)0.43×10^8 Hz (D)0.43 GHz

Q.17 **Assertion (A):** A transmission line supports TEM mode whereas the waveguides cannot support TEM modes.

Reason (R): TEM mode can exist only where there are two conductors and cross-sectional dimensions are large in comparison to the operating wavelength.

(A)Both (A) and (R) are true and (R) is the correct explanation of (A).

(B)Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C)Statement (A) is true but (R) is false.

(D)Statement (A) is false but (R) is true.

Q.18 A parallel-plate waveguide is filled with air and the distance between the plates is 40 mm. At a frequency of 5 GHz, the mode(s) that can propagate in the waveguide is/are-

(A)TM_0 (B)TM_1

(C)TE_1 (D)TM_2

Q.19 Consider a circular waveguide that has d as its cross-sectional diameter. Which one of the following options is correct for the ratio of cut-off frequencies of TE and TM mode of operation to be least?

(A)It is dependent on the root of Bessel harmonics.

(B)It is dependent on the derivatives of Bessel harmonics.

(C)It is dependent on the derivatives of Bessel harmonics & value is less than unity.

(D)It is dependent only on Bessel harmonics.

Q.20 An a × b rectangular waveguide is operating in four different modes as $TM_{11}, TM_{12}, TM_{10}$, and TE_{20}. If a = 2b then the ascending order of the operating modes for their cut-off frequencies will be

(A)$TE_{10} < TE_{20} < TM_{11} < TM_{12}$
(B)$TE_{10} < TE_{11} < TM_{20} < TM_{12}$

(C)$TE_{12} < TE_{11} < TM_{20} < TM_{10}$
(D)$TE_{10} < TE_{20} < TM_{12} < TM_{11}$

Q.21 Which of the following statement(s) is/are **not true**?

(A)The TE_{10} mode has highest attenuation among all modes in a rectangular waveguide.

(B)In a waveguide: Guided wavelength < Intrinsic wavelength

(C) Group velocity $= \left[\frac{\partial \beta}{\partial \omega}\right]^{-1}$

(D)For infinite length transmission line, input impedance < Z_0.

Q.22 An air filled rectangular waveguide with dimension of 8cm × 5cm operates in the TM_{23} mode at 10GHz. The characteristic wave impedance is

(A)$26.66\pi\Omega$ (B)$44.62\pi\Omega$

(C)$89.93\pi\Omega$ (D)$540\pi\Omega$

Q.23 An air filled rectangular waveguide having inner dimensions of 4cm × 3cm. If waveguide is operating at 15GHz, then the wave impedance of TE_{20} mode of wave propagation in the waveguide is _____Ω. (Up to 2 decimal places)

Q.24 The magnetic field of the TE_{11} mode of the rectangular waveguide having a × b as dimension is defined

as $H_z = H_0 \cos(0.3\pi x) \cos(0.5\pi y)$, where x and y are in cm.

The cut-off frequency for this case is _____(in GHz).

Q.25 Evaluate the peak value of the electric field (in V/m and up to three decimal places) in an air-filled waveguide of dimensions of 4.8 cm × 2.2 cm. The waveguide transports the energy of 4mW in the dominant mode and the frequency rate of operation is 12 GHz.

Q.26 A TM mode wave operating at 15 GHz is propagated in an air-filled rectangular wave guide. If a field component of the wave is given as,

$$E_z = \sin\left(\frac{4\pi x}{a}\right)\sin\left(\frac{2\pi y}{b}\right)\cos(\omega t - 80\pi z)\ \text{V/m}$$

Then cutoff frequency (in GHz) of the operating mode is _____. (Rounded off to the nearest integer)

Answer Key

Q.1 Ans. [4.8]

Q.2 Ans. [2.74]

Q.3 Ans. [0]

Q.4 Ans. [B]

Q.5 Ans. [D]

Q.6 Ans. [65.99]

Q.7 Ans. [4.395]

Q.8 Ans. [2.6]

Q.9 Ans. [B]

Q.10 Ans. [A]

Q.11 Ans. [C]

Q.12 Ans. [B]

Q.13 Ans. [D]

Q.14 Ans. [A]

Q.15 Ans. [A]

Q.16 Ans. [A]

Q.17 Ans. [C]

Q.18 Ans. [A, B, C]

Q.19 Ans. [D]

Q.20 Ans. [A]

Q.21 Ans. [A, B, D]

Q.22 Ans. [A]

Q.23 Ans. [435.32]

Q.24 Ans. [8.7]

Q.25 Ans. [76.916]

Q.26 Ans. [9]

6 Antennas

MCQ, MSQ and NAT

Q.1 The radiated power from an antenna whose radiation resistance is 400 Ω operating at a frequency of 2 GHz and with a current of 3.5 Amperes is _____W. [Rounded off up to the nearest integer].

Q.2 The effective area of a Hertzian dipole antenna operating at 125 MHz is_____m^2. (Rounded off up to two decimal places).
{Consider directivity of Hertzian dipole as 1.5}

Q.3 For a half-wave dipole antenna, if the directivity is 1.64, ohmic resistance is 0.8 Ω and the effective area is 0.047 m^2, then-
[Consider radiation resistance as 73 Ω]

(A)Operating frequency= 250 MHz

(B)Efficiency= 98.91%

(C)Operating frequency= 500 MHz

(D) Efficiency= 49.45%

Q.4 Calculate the Null-to-Null beam width (BWFN) of a broadside array, if the number of elements is 100 and the spacing between the elements is $\frac{3\lambda}{4}$.

(A)0.401 radians.

(B)0.026 radians.

(C)0.807 radians.

(D)0.046 radians.

Q.5 The radiation resistance of a current carrying element whose overall length is $\frac{\lambda}{60}$ is-

(A)0.340 Ω

(B)0.110 Ω

(C)0.219 Ω

(D)0.475 Ω

Q.6 Consider that the transmitting and receiving antennas are separated by

250 λ distance and the directive gains are 316.23 and 63.1 respectively. If the received power is 5 mW, then the minimum transmitted power will be_____watts. (Rounded off up to three decimal places).

Q.7 Consider a radar antenna operating at 2.5 GHz and radiating power of 300 kW. The effective aperture is 10 m^2. What will be the received power by an antenna, if the target radar cross-section is 30 m^2 and is at a distance of 500 km?

(A)57.15×10^{16} W

(B)57.15×10^{-16} W

(C)79.57×10^{15} W

(D)79.57×10^{-15} W

Q.8 The Polarization loss factor -

(A)is the same as power dissipation.

(B)occurs due to polarization mismatch.

(C)causes loss of EM power.

(D)At no polarization mismatch, the value of PLF is 1.

Q.9 The directivity of an antenna with circular aperture of diameter 4 meter at frequency of 7 GHz is_____. (Rounded off to the nearest integer).

[Consider that the aperture efficiency or aperture coefficient is 0.85]

Q.10 A radiating antenna operating at 20 MHz has a radiation resistance of 50 Ω. It is fed with an r.m.s. current of 15 A and the bandwidth is 400 kHz. Which of the following is correct?

A)The quality factor is 20.

(B)The power radiated is 11.25 W.

(C)The quality factor is 50.

(D)The power radiated is 11.25 kW

Q.11 Consider an isotropic antenna radiating in free space. At a distance of 110 meters from the antenna, the total electric field (E_0) is measured to be 10 V/m. The power radiated will be_____kW. [Up to two decimal places].

Q.12 Which of the following statements given in options regarding arrays is/are correct?

(A)In a broadside array, the maximum radiation occurs in a direction perpendicular to the array axis.

(B)The directivity of the end-fire antenna array is described as $\frac{4nd}{\lambda}$.

(C)Broadside arrays are used for transmission purposes.

(D)End-fire arrays are used for reception.

Q.13 A Hertzian dipole antenna of length $\lambda/4$ is having total loss resistance of 2Ω. The antenna efficiency is-

(A)88.7%

(B)96.1%

(C)68.0%

(D)72.5%

Q.14 A parabolic antenna is having a power gain of 1600 at the wavelength of 0.6 m, then the half-power beamwidth (in degree) is_____ (Up to 2 decimal places)

Q.15 A reflector antenna used for a communication base station operates at 19 GHz with a power gain of 40 dB, the radiation efficiency of 98 %, and diameter of 0.2 m. The effective aperture of the antenna is_____ m^2. (Up to 1 decimal place)

Q.16 A Hertzian dipole antenna of length 0.02 m is operating at a frequency of 90 MHz. To radiate 110 W of power the peak value of the required current is _____ A.

Q.17 Consider the following statements about an antenna.

1. It is a transducer
2. Its performance is essentially frequency sensitive.
3. It is a reciprocal device.
4. With an increase in its effective aperture area, width of the radiated beam increases.

Which of the above statements are correct?

(A)1, 2 and 3

(B)1, 2 and 4

(C) 2, 3 and 4 only

(D)1, 2, 3 and 4 only

Q.18 A Yagi-Uda antenna has a driven antenna with-

(A) Only directors

(B) only a reflector

(C) only one or more directors

(D) a reflector and one or more directors

Q.19 The directivity of the loop antenna having 0.5 m radius and operating at 1.2 MHz frequency is -

(A)3

(B)0.012

(C)1

(D)1.5

Q.20 The effective area of a Hertzian dipole antenna which is operating at 120 MHz of frequency will be -

(A)$0.74\ \mathrm{m}^2$

(B)$1.74\ \mathrm{m}^2$

(C)$1.5\ \mathrm{m}^2$

(D)$2.75\ \mathrm{m}^2$

Q.21 The electric field intensity (in V/m) at a distance of 5 km form an antenna having a directive gain of 5 dB and radiating a total power of 10 kW is-

(A)0.194

(B)0.094

(C)0.075

(D)0.574

Q.22 Choose the correct statements for an antenna field-

1. Far field is represented by $\frac{1}{r^3}$
2. Induction field is represented by $\frac{1}{r^2}$
3. Electrostatic field is represented by $\frac{1}{r}$
4. Far field is represented by $\frac{1}{r}$

(A)1, 2 and 3 only

(B)2 and 4 only

(C)1 and 3 only

(D)3 and 4 only

Q.23 If a loop antenna with loop radius 0.8 m is made up of copper wire of radius 8 mm ,the loop radiates at 7.5 MHz, carries current 30A and conductivity of 6×10^7 S. The radiation resistance, radiation power and radiation efficiency will be, respectively -

(A)$49\ \mathrm{m\Omega}, 22.1\ \mathrm{W}$ and 81.4%

(B)$20\ \mathrm{m\Omega}, 10\ \mathrm{W}$ and 37%

(C)$25\ \text{m}\Omega, 10.15\ \text{W}$ and 27%

(D)$24.4\ \text{m}\Omega, 9.5\ \text{W}$ and 70%

Q.24 If an antenna in free space receives the power of 6μW when the incident RMS electric field is 30 mV/meter, then the effective aperture (in m^2) of the antenna is -

(A)3

(B)1.25

(C)5

(D)2.52

Q.25 The current in a short circuit element of length $l = 0.03\lambda$ is given by

$$I(z) = \left\{ \begin{array}{ll} \frac{I_0}{2} & for 0 < |z| \leq \frac{1}{4} \\ I_0 & for \frac{1}{4} < |z| \leq \frac{1}{2} \end{array} \right\}$$

What will be the radiation resistance (in Ω) of the element____ (Answer up to 1 decimal)?

Q.26 Consider a parabolic dish antenna with conical beam 5° wide. The directivity of the antenna is _____ (in dB).

(A)1650.12

(B)16

(C)32.17

(D)36.5

Q.27 The radiation intensity of an antenna is given by-

$$U(\theta) = \cos^3\theta; \quad 0 \leq \theta \leq \frac{\pi}{2}$$

The directivity of antenna is-

(A)18.06 dB

(B)9.03 dB

(C)12.05 dB

(D)24.1 dB

Q.28 Consider that the two Hertzian dipoles are placed orthogonal to each other and are fed by currents of equal amplitude with a phase difference of 90°.

The far electric field is given as –

$$\vec{E} = \frac{j\beta\eta(Idl)}{4\pi r} e^{-j\beta r} \left[(\sin\theta - j\cos\theta\cos\phi)\hat{a}_\theta + (j\sin\phi)\hat{a}_\phi\right]$$

The far zone magnetic field intensity and maximum directivity will be -

(A) $\frac{\eta(\beta Idl)^2}{32\pi^2}\left(1 + \sin^2\phi\sin^2\theta\right)$

(B) $\frac{j\beta(Idl)}{4\pi r} e^{-j\beta r}\sin\phi$

(C) $\frac{j(\beta Idl)}{4\pi r} e^{-j\beta r}(\sin\theta - j\cos\theta\cos\phi)$

(D) $\frac{3}{2}$

Q.29 Maximum electric field strength radiated by an antenna is 6mV/m measured at 40 km from the antenna. If the antenna radiates a total power of 100 kW, then the directivity of antenna is

(A)-2.02 dB

(B)9.6 dB

(C) 0.0096 dB

(D)-20 .18 dB

Q.30 The field Radiation pattern of an antenna is given as

$$F(\theta) = \begin{cases} \cos^4\theta & 0 \le \theta \le \pi/2, 0 \le \phi \le 2\pi \\ 0 & elsewhere \end{cases}$$

Directivity of antenna is dB, (correct up to two decimal points)

Answer Key

Q.1 Ans. [4900]

Q.2 Ans. [0.69]

Q.3 Ans. [B, C]

Q.4 Ans. [B]

Q.5 Ans. [C]

Q.6 Ans. [2.473]

Q.7 Ans. [D]

Q.8 Ans. [B, C, D]

Q.9 Ans. [72594]

Q.10 Ans. [C, D]

Q.11 Ans. [20.16]

Q.12 Ans. [A, B, C]

Q.13 Ans. [B]

Q.14 Ans. [4.29]

Q.15 Ans. [0.2]

Q.16 Ans. [88.62]

Q.17 Ans. [A]

Q.18 Ans. [D]

Q.19 Ans. [D]

Q.20 ` Ans. [A]

Q.21 Ans. [C]

Q.22 Ans. [B]

Q.23 Ans. [A]

Q.24 Ans. [D]

Q.25 Ans. [0.4]

Q.26 Ans. [C]

Q.27 Ans. [B]

Q.28 Ans. [B, C, D]

Q.29 Ans. [D]

Q.30 Ans. [12.55]

7 Optical Fibres

MCQ, MSQ and NAT

Q.1 Consider a silica optical fiber with a core diameter large enough to be considered by ray theory analysis that has a core refractive index of 1.50 and a cladding refractive index of 1.47. The acceptance angle in the air for the fiber will be_____ degree. (Rounded off up to two decimal places)

Q.2 If an optical fiber in the air has a 0.4 numerical aperture, then the difference between the acceptance angle for meridional rays and the skew rays which change the direction by 100° at each reflection is _____degrees. (Rounded off up to two decimal places)

Q.3 Determine the refractive index of cladding in an optical fiber with a critical angle of 80° and the refractive index of the core is 1.45.

(A)2.01 (B)1.85

(C)1.11 (D)1.42

Q.4 A graded-index fiber with a parabolic refractive index profile core has a refractive index of 1.5 and the relative index difference is 1%. The maximum possible core diameter is _____ $(\mu \mathrm{m})$ which allows single-mode operation at a wavelength of 1.3 $\mu \mathrm{m}$. (Rounded off up to two decimal places)

Q.5 A step-index multimode optical fiber has a diameter of 7.5 µm and the refractive indices of core and cladding are 1.49 and 1.3 respectively. The number of modes(approximately) that can propagate into the fiber if the wavelength of the source is 2.5 µm is _____(Rounded off up to the nearest integer)

Q.6 The critical angle (in degrees) when the core refractive index is 1.35 and the relative refractive index difference is 3 % is-

(A)151.86° (B)57.93°

(C)97.93° (D)75.93°

Q.7 The maximum acceptance angle for a fiber having core refractive index of 1.45 and cladding refractive index of 1.4 is _____.

(A)11.09° (B)22.18°

(C)44.36° (D)30°

Q.8 **Assertion(A):** Total internal reflection is only happened if core refractive index is less than cladding refractive index.

Reason(R): Internal dispersion is arising due to presence of multiple path for light transmission in fiber.

(A)Both A and R are true, and R is the correct explanation of A.

(B)Both A and R are true, but R is not correct explanation of A.

(C)A is true but R is false.

(D)A is false but R is true.

Q.9 During an experiment, optical fiber used has a critical angle of 60. What will be the numerical aperture for the given optical fiber.

(A) 0.763 (B) 0.5

(C) 0.707 (D) 0.867

Q.10 The number of subscribers an optical fiber communication system working on a wavelength of 1.2 mm can feed, if a channel requires 10 kHz is-

(A) 1.15×10^{10}

(B)10^{10}

(C)2.3×10^{10}

(D)2.5×10^{10}

Q.11 Consider the following statements – The radiation losses are caused mainly by -

(1) Impurities in the fiber material.

(2) Bending of the fiber by geometric imperfections.

(3) Constant-radius bends.

(4) Thermal contractions between cladding and core materials.

Which of these statements are correct?

(A) 1, 2 and 3.

(B) 1, 3 and 4.

(C) 1, 2 and 4.

(D) 2, 3 and 4.

Q.12 The multipath time dispersion of a glass fiber with refractive indices of core and cladding as 1.5 and 1 will be-

(A) 2.5 ns/m

(B) 3.5 ns/m

(C) 4 ns/m

(D) 5 ns/m

Q.13 Light pulses propagate through a fiber cable with an attenuation of 0.25 dB/km. Determine the distance (in km) through which the power of pulses is reduced by 60%.

(A) 1591 (B) 159.1

(C) 1.591 (D) 15.91

Q.14 An optical fiber has V-parameter or Normalized frequency as 10. Find the number of modes supported by the fiber and percentage power in the cladding respectively.

(A)50 and 12.45

(B)100 and 20.42

(C)100 and 16.32

(D)50 and 18.86

Q.15 A step-index fiber has a core refractive index of 1.56 and a

numerical aperture of 0.19. If the core diameter is 70 µm, then the number of modes that the fiber can propagate at a wavelength of 0.7 µm is_____. (Rounded off to the nearest integer).

Answer Key

Q.1	Ans.	[17.37]
Q.2	Ans.	[14.91]
Q.3	Ans.	[D]
Q.4	Ans.	[4.68]
Q.5	Ans.	[23]
Q.6	Ans.	[D]
Q.7	Ans.	[B]
Q.8	Ans.	[D]
Q.9	Ans.	[D]
Q.10	Ans.	[D]
Q.11	Ans.	[D]
Q.12	Ans.	[A]
Q.13	Ans.	[D]
Q.14	Ans.	[D]
Q.15	Ans.	[1781]